Emergent Firm Heterogeneity Through Learning From Experience and Costly Deliberation

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Abstract

We adapt Ilut and Valchev [2022] to a firm investment setting in which firms face frictions in learning in a stochastic environment. Firms are modeled as reinforcement learning agents who learn from experience (*model-free* learning) which resembles temporal difference learning, and from planning (*model-based* learning). Firms develop investment plans based on imperfect learning from past experience and costly deliberation. Each firm resides in a stochastic environment with idiosyncratic productivity shocks which shape their policy formation. We show that this framework generates a heterogeneous ergodic distribution over firm beliefs and capital stocks. The resulting firm behavior is consistent with several empirical regularities that have individually motivated distinct literatures including lumpy investment dynamics, persistent dispersion in marginal products of capital across observably similar firms, sluggish aggregate investment response to identified shocks, and history-dependent managerial decision-making.

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1 Introduction, Literature Review, and Summary

In canonical models of firm investment, firms are perfectly rational agents capable of solving infinite-horizon dynamic optimization problems. The firm uses capital $k_{j,t}$ (physical machines, equipment, and infrastructure) to produce output via a production function $f(z_{j,t}, k_{j,t})$, where $z_{j,t}$ is a stochastic productivity shock that captures fluctuations in the firm's efficiency (e.g., demand shifts, supply disruptions, technological changes). Each period, the firm can purchase new capital by investing $i_{j,t}$, but the firm incurs convex adjustment costs $\psi(i_{j,t}, k_{j,t})$ that penalize rapid changes to its capital stock (frictions of retooling factories, training workers on new equipment, and reorganizing production). Capital depreciates over time at rate δ (machines wear out, equipment becomes obsolete), so the firm's capital stock evolves as $k_{j,t+1} = (1-\delta)k_{j,t} + i_{j,t}$: what survives from today plus what is installed. To finance investment, the firm can issue bonds $b_{j,t+1}$ (borrow from external creditors), but lenders require collateral: borrowing is capped at a fraction θ of next-period capital.

This problem admits an analytical closed-form solution. Under this so-called neoclassical model, the optimal investment rule is simple: invest until the shadow value of an additional unit of capital (Tobin's q) equals its installation cost. When $q > 1$, a new machine generates more value than it costs to install, so the firm invests; when $q < 1$, existing machines are worth less than their replacement cost, so the firm divests.

While elegant, there are well-documented empirical irregularities that pose challenges to this model.

1. **Weak investment-q relationship.** The neoclassical model predicts that Tobin's q is a sufficient statistic for investment. Empirically, q explains almost none of the variation in firm-level investment. [Fazzari et al., 1988, Hubbard, 1998, Erickson and Whited, 2000, Abel and Eberly, 1996].

2. **Excess sensitivity to cash flow.** The neoclassical model predicts investment depends only on q, not on internal funds. Empirically, investment responds strongly to cash flow even after controlling for q. [Fazzari et al., 1988, Kaplan and Zingales, 1997, Lamont, 1997, Abel and Eberly, 1996].

3. **Excess investment dispersion.** The neoclassical model predicts firms with identical fundamentals make identical choices. Empirically, firms with similar observables make persistently different investment decisions. [Hsieh and Klenow, 2009, Bachmann and Bayer, 2014]

4. **Investment inertia and lumpy adjustment.** The neoclassical model with convex costs predicts smooth, continuous adjustment. Empirically, firms underreact for extended periods then make large discrete changes. [Doms and Dunne, 1998, Cooper and Haltiwanger, 2006].

5. **Asymmetric investment dynamics.** The neoclassical model predicts symmetric responses to positive and negative shocks. Empirically, firms expand gradually but

contract sharply.[Abel and Eberly, 1996, Caballero and Engel, 1999].

6. **Fat-tailed firm size distribution.** The neoclassical model predicts convergence to a common steady state. Empirically, firm sizes exhibit persistent heterogeneity with a fat right tail. [Lucas, 1978, Luttmer, 2007]

This project proposes to address these puzzles by adapting the framework of Ilut and Valchev [2022][1] to the context of firm investment. In this model, agents do not know the true value of different actions. They may hold some initial beliefs (a prior) over the action-value function $Q^*(s, a)$, which encodes the present discounted value of taking action a in state s and behaving optimally thereafter, but they must learn and refine these beliefs over time through experience and reasoning (planning). From the firm's perspective, the true action-value function Q^* is unobserved, and the firm's beliefs $(\hat{Q}_t, \hat{\Sigma}_t)$ about it become part of the relevant state.

The firm operates as an RL agent that explores the state-action space to learn about Q^*, it uses temporal-difference learning (*model-free* RL) to update its beliefs as well as some planning (or reasoning) (*model-based* RL) to refine them, and it ultimately develops an investment policy to maximize the expected present value of dividends. The true underlying MDP is, in fact, the neoclassical model described above, so the true Q^* is the known solution to that problem. However, firms in exploring and learning about Q^* may generate a dispersion of investment policies that are path dependent as they are each subject to idiosyncratic shocks. This dispersion of policies may be persistent and not perfectly aligned with the neoclassical solution. In this framework we generate an ergodic distribution of firm-level investment policies that can match the empirical puzzles.

2 Investment Problem

We consider the canonical dynamic investment problem of a single firm (firm j) that serves as the standard model of firm-level capital allocation in macroeconomics. The firm operates a production function $f(z_{j,t}, k_{j,t})$, concave and increasing in both arguments, where $z_{j,t}$ is an idiosyncratic productivity shock governing the firm's current efficiency and $k_{j,t}$ is its capital stock, the physical machinery and equipment at its disposal.

The firm enters the period with a stock of debt $b_{j,t}$ which carries a risk-free interest rate r. Each period, the firm earns a dividend:

$$d_{j,t} = f(z_{j,t}, k_{j,t}) - i_{j,t} - \psi(i_{j,t}, k_{j,t}) + b_{j,t+1} - (1 + r)b_{j,t}. \tag{1}$$

This represents the residual of output after investment expenditure $i_{j,t}$, capital adjustment costs $\psi(i_{j,t}, k_{j,t})$, and the net cash flow from issuing new debt $b_{j,t+1}$ and repaying existing debt obligations $(1 + r)b_{j,t}$. The firm derives a per-period payoff $u(d_{j,t})$ from this dividend. The firm's problem is to choose policies for investment and debt that maximize the expected

[1] Ilut and Valchev [2022] develop their framework in the context of a consumption-savings problem. This project extends it to the richer and empirically distinct setting of firm investment.

present discounted value of its payoff stream. It is common to model firms as risk-neutral agents with a linear payoff function, $u(d_{j,t}) = d_{j,t}$.

$$\max_{\{i_{j,t},\, b_{j,t+1}\}_{t=0}^{\infty}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t d_{j,t} \right] \tag{2}$$

The firm faces several frictions. First, a capital adjustment cost $\psi(i_{j,t}, k_{j,t})$ penalizes rapid changes in the capital stock; installing or dismantling capital is costly and disruptive. We consider a standard quadratic adjustment cost,

$$\psi(i_{j,t}, k_{j,t}) = \frac{\phi}{2} \left(\frac{i_{j,t}}{k_{j,t}} \right)^2 k_{j,t}, \tag{3}$$

where $\phi \geq 0$. This cost is convex in the rate of investment; the firm cannot freely jump to its desired capital stock and must smoothly adjust its capital stock. Second, a collateral borrowing constraint limits the firm's ability to finance investment externally. The firm can only issue new debt $b_{j,t+1}$ up to a fraction θ of its next-period capital stock:

$$b_{j,t+1} \leq \theta k_{j,t+1}. \tag{4}$$

This bounds the feasible investment action at each state and introduces the possibility that profitable investment opportunities go unexploited due to insufficient internal funds.

The capital stock depreciates over time at a per-period rate δ, evolving according to a standard law of motion:

$$k_{j,t+1} = (1 - \delta) k_{j,t} + i_{j,t}. \tag{5}$$

The firm's problem is therefore a dynamic programming problem with continuous state and action spaces. Defining the state as the firm's idiosyncratic productivity, capital, and debt $(z_{j,t}, k_{j,t}, b_{j,t})$, the problem can be expressed as the recursive Bellman equation:

$$V(z_{j,t}, k_{j,t}, b_{j,t}) = \max_{i_{j,t},\, b_{j,t+1}} \{ d_{j,t} + \beta\, \mathbb{E}[V(z_{j,t+1}, k_{j,t+1}, b_{j,t+1}) \,|\, z_{j,t}] \} \tag{6}$$

subject to:

$$d_{j,t} = f(z_{j,t}, k_{j,t}) - i_{j,t} - \frac{\phi}{2} \left(\frac{i_{j,t}}{k_{j,t}} \right)^2 k_{j,t} + b_{j,t+1} - (1 + r) b_{j,t} \tag{7}$$

$$k_{j,t+1} = (1 - \delta) k_{j,t} + i_{j,t} \tag{8}$$

$$b_{j,t+1} \leq \theta k_{j,t+1} \tag{9}$$

$$d_{j,t} \geq 0 \tag{10}$$

where $V(\cdot)$ is the value function and the expectation is taken over the stochastic evolution of productivity conditional on the current state.

2.1 Solution to Canonical Investment Problem

The firm's optimal investment policy can be summarized using the shadow value of installed capital ($q_{j,t}$). When the collateral constraint is slack, the optimal investment-capital ratio is linear in $(q_{j,t} - 1)$, where $q_{j,t}$ denotes the marginal value of installed capital. When the collateral constraint binds, however, investment is limited by the firm's borrowing capacity and internal funds. The resulting optimal investment policy can therefore be written as the following.

$$\frac{i_{j,t}}{k_{j,t}} = \begin{cases} \frac{1}{\phi}(q_{j,t} - 1), & \text{if } b_{j,t+1} < \theta k_{j,t+1} \\ \frac{k^{BC}_{j,t+1} - (1-\delta)k_{j,t}}{k_{j,t}}, & \text{if } b_{j,t+1} = \theta k_{j,t+1} \end{cases} \tag{11}$$

The term $k^{BC}_{j,t+1}$ denotes the level of next-period capital implied by the binding borrowing constraint and the firm's budget constraint. Intuitively, when the value of an additional unit of capital exceeds its installation cost, the firm increases investment. When the collateral constraint binds, capital also has value because it relaxes the borrowing limit, and investment becomes limited by available collateral rather than the marginal q condition. The full derivation of the firm's first-order conditions and the resulting Tobin's q investment rule is provided in Appendix A.1.

2.2 The Firm's Problem and Policy Under Specific Functional Forms

To this point, we have specified the firm's investment problem and derived the firm's optimal investment policy in general terms. For simplicity, we choose production to be Cobb-Douglas, $f(z_{j,t}, k_{j,t}) = z_{j,t}k^{\alpha}_{j,t}$, and the productivity shock to follow an AR(1) process, $\ln z_{j,t+1} = \rho \ln z_{j,t} + \varepsilon_{j,t+1}$ with $\varepsilon_{j,t+1} \sim \mathcal{N}(0, \sigma^2_z)$. We also assume linear utility, $u(d_{j,t}) = d_{j,t}$, so that the firm is risk-neutral. Under these assumptions, the Bellman equation characterizing the firm's problem takes the following explicit form, subject to the capital accumulation law of motion and the collateral constraint, as described previously.

$$\begin{aligned} V(z_{j,t}, k_{j,t}, b_{j,t}) = \max_{i_{j,t},\, b_{j,t+1}} \Bigg\{ & z_{j,t}k^{\alpha}_{j,t} - i_{j,t} - \frac{\phi}{2}\left(\frac{i_{j,t}}{k_{j,t}}\right)^2 k_{j,t} \\ & + b_{j,t+1} - (1+r)b_{j,t} + \beta\,\mathbb{E}\left[V(z_{j,t+1}, k_{j,t+1}, b_{j,t+1}) \,|\, z_{j,t}\right] \Bigg\} \end{aligned} \tag{12}$$

Intuitively, the value function represents the fundamental value of the firm given its current state (defined by its productivity, capital, and debt $(z_{j,t}, k_{j,t}, b_{j,t})$) under a specific set of policies. It decomposes this value into two parts: the current dividend and the discounted expected value of future payoffs. The current dividend is simply the firm's production revenue, $z_{j,t}k^{\alpha}_{j,t}$, minus the total cost of investment including adjustment frictions, $i_{j,t} + \frac{\phi}{2}\left(\frac{i_{j,t}}{k_{j,t}}\right)^2 k_{j,t}$, plus the net cash flow from debt management, $b_{j,t+1} - (1+r)b_{j,t}$. Consequently, the maximized value function, $V(\cdot)$, corresponds to the fundamental equity value of the firm when it optimally executes its investment and financing decisions. Because the dividend represents the residual cash flow to owners after all operational and debt obligations are settled, maximizing this value function is economically equivalent to maximizing shareholder value.

Taking the first-order condition with respect to investment, the optimal policy relies on the shadow value of installed capital, $q_{j,t}$, which is explicitly defined as the expected marginal continuation value of an additional unit of capital:

$$q_{j,t} = \beta\,\mathbb{E}\left[\frac{\partial V(z_{j,t+1}, k_{j,t+1}, b_{j,t+1})}{\partial k_{j,t+1}} \,\middle|\, z_{j,t}\right]. \tag{13}$$

Using this definition, the firm's optimal investment-to-capital ratio can be written as:

$$\frac{i_{j,t}}{k_{j,t}} = \begin{cases} \frac{1}{\phi}(q_{j,t} - 1), & \text{if } b_{j,t+1} < \theta k_{j,t+1} \\ \frac{k^{BC}_{j,t+1} - (1-\delta)k_{j,t}}{k_{j,t}}, & \text{if } b_{j,t+1} = \theta k_{j,t+1} \end{cases} \tag{14}$$

To visualize the rational firm's optimal policy, we show canonical phase diagrams in Fig. 1: the optimal next-period capital stock k_{t+1} as a function of current capital k_t for different productivity levels z_t, and the corresponding marginal value of capital q_t[2] as a function of k_t. Together these characterize the firm's policy: when $q_t > 1$ the firm invests and grows its capital stock; when $q_t < 1$ it disinvests. The firm invests such that $k_{t+1} = i_t + (1-\delta)k_t$ and $q_t = 1$ when the collateral constraint is slack, and up to the borrowing limit when it binds. Because the rational firm knows the optimal policy under any state (z_t, k_t, b_t), it almost always avoids a binding collateral constraint. A large negative productivity shock followed by a large positive shock can produce a binding constraint, but this occurs with low probability under the ergodic distribution.

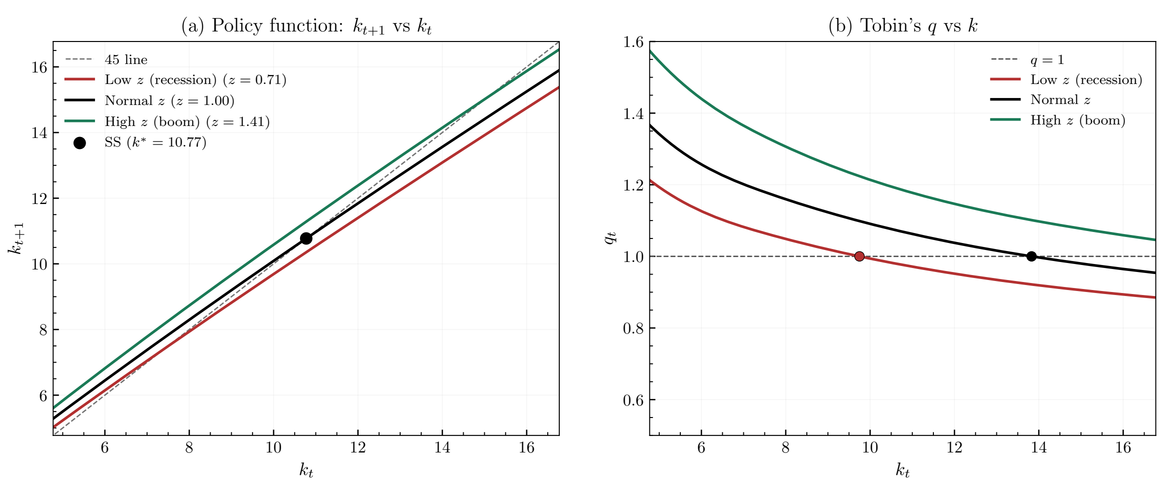

Figure 1: Phase diagrams of the rational firm's optimal policy.

Notice in Fig. 1 that the $q = 1$ crossings in panel (b) do not align with the steady-state

[2] Equivalently, the shadow price of capital or the Lagrange multiplier on the capital accumulation constraint.

crossings in panel (a). This is because $q = 1$ marks the extensive margin of investment, the capital stock at which the firm is indifferent between adding and not adding a unit of capital, i.e. $i = 0$. The 45 crossing marks the intensive margin, the capital stock at which the firm invests exactly enough to offset depreciation, $i = \delta k$. At the $q = 1$ crossing, the firm switches from divestment ($i < 0$, allowing capital to decay) to active investment ($i > 0$). At the 45 crossing, the firm reaches its stationary capital stock where gross investment equals depreciation. The two are separated by the adjustment cost wedge: at steady state, $q^* = 1 + \kappa\delta > 1$, so the firm must value capital strictly above replacement cost to justify the flow cost of maintaining it.

2.3 Investment Problem Under Learning Through Reasoning and Experience

With the canonical formulation in hand, we can now transition to the RL framework. We now write the firm's problem using the reasoning and experience framework described in Appendix B. Importantly, the firm exists in the environment described previously (Section 2), but it does not know the solution (Section 2.1). The firm attempts to learn about its optimal policy through two processes: experience and planning.

The experience channel is a temporal-difference learning process with a Bayesian update governed by a Kalman gain, formally described in Definition 6. Because the environment is stochastic, the firm may misattribute outcomes to actions. For example, a particularly adverse productivity realization can cause the firm to infer that its investment was too aggressive, while a favorable shock can reinforce excessive divestment. The reasoning channel resembles a model-based reinforcement learning process. At a cognitive cost, the firm mentally simulates candidate actions and their consequences, refining its beliefs about the action-value function beyond what experience alone provides. This is formally described in Definition 7.

The two channels are linked through a certainty equivalence assumption: the firm chooses its reasoning intensity under the expected realization of the reasoning signals, so that reasoning reduces the posterior variance of Q^* but does not shift the conditional mean. The cognitive cost of reasoning is proportional to the entropy reduction achieved, weighted by the shadow price of the entropy constraint (the Lagrange multiplier on the exploration bound). This shadow price captures the opportunity cost of uncertainty: when forced exploration is costly, variance reduction is worth the cognitive expenditure. Because the reasoning step preserves the conditional mean by construction, the belief sequence $\{\hat{Q}_t\}$ is a martingale with respect to the reasoning filtration. The firm cannot expect its own future reasoning to correct current errors, so deviations from the rational policy are persistent.

At each time t, the firm exists in state defined by its capital stock k_t, debt b_t, and productivity z_t ($\mathbf{s}_t = (z_t, k_t, b_t)$). The firm chooses both investment action i_t and borrowing action b_{t+1} ($\mathbf{a}_t = (i_t, b_{t+1})$) which determine the next-period state through the transition function

$\mathcal{F}(i_t, b_{t+1}, (z_t, k_t, b_t), \varepsilon_{t+1})$.

$$\mathbf{s}_{t+1} = \begin{bmatrix} z_{t+1} \\ k_{t+1} \\ b_{t+1} \end{bmatrix} = \mathcal{F}\left((i_t, b_{t+1}), (z_t, k_t, b_t), \varepsilon_{t+1}\right) = \begin{bmatrix} \exp\left(\rho \ln z_t + \varepsilon_{t+1}\right) \\ (1-\delta)k_t + i_t \\ b_{t+1} \end{bmatrix} \tag{15}$$

The optimal policy is retrieved by solving the firm's constrained optimization problem conditional on the state $\mathbf{s}_t = (z_t, k_t, b_t)$. In this context, the *Value Function* (V) represents the maximized fundamental equity value of the firm, and the *Action-Value Function* (Q) characterizes the immediate dividend from a specific investment and financing choice plus the expected discounted future value.

Proposition 1 (Optimal Policy and Value Functions)**.** *The optimal policy π^* maps the state $\mathbf{s}_t = (z_t, k_t, b_t)$ to actions $\mathbf{a}_t = (i_t, b_{t+1})$ subject to the collateral borrowing constraint. The Action-Value Function Q^* and Value Function V^* satisfy:*

$$Q^*(\mathbf{s}_t, \mathbf{a}_t) = z_t k_t^\alpha - i_t - \frac{\phi}{2}\left(\frac{i_t}{k_t}\right)^2 k_t + b_{t+1} - (1+r)b_t + \beta \mathbb{E}\left[V^*(z_{t+1}, k_{t+1}, b_{t+1}) \mid z_t\right] \tag{16}$$

$$V^*(\mathbf{s}_t) = \max_{\mathbf{a}_t \in \Gamma(\mathbf{s}_t)} Q^*(\mathbf{s}_t, \mathbf{a}_t) = Q^*(\mathbf{s}_t, \pi^*(\mathbf{s}_t)) \tag{17}$$

where $\Gamma(\mathbf{s}_t)$ represents the feasible action space defined by the collateral constraint $b_{t+1} \leq \theta((1-\delta)k_t + i_t)$ and the non-negative dividend condition $d_t \geq 0$.

The firm is Bayesian and has beliefs over the true action-value function Q^* described by a Gaussian Process with mean $\hat{Q}_t$ and covariance $\hat{\Sigma}_t$, identical to Appendix B.1. As described in Appendix B.7, the firm's *true* problem is a Bayes-adaptive Markov decision process (BAMDP) where the true Q^* is unknown and beliefs continuously evolve. We assume the firm cannot solve it perfectly.[3] Instead, the firm approximates the optimal policy over time through imperfect reasoning and experience. The experience and reasoning cycle each period as described in Definition 2.

Proposition 2 (Investment as the Sole Dynamic Choice Variable)**.** *The firm's state can be reduced from $\mathbf{s}_t = (z_t, k_t, b_t)$ to $\mathbf{s}_t = (z_t, k_t)$ by substituting two case optimal borrowing policy. The reduced Bellman equation is:*

$$\begin{aligned} V^*(z_t, k_t) = \max_{i_t} \Big\{ & f(z_t, k_t) - i_t - \psi(i_t, k_t) + \theta k_{t+1}\big[1 - \beta(1+R)\big] \\ & + \beta\, \mathbb{E}[V^*(z_{t+1}, k_{t+1}) \mid z_t] \Big\} \end{aligned} \tag{18}$$

where $k_{t+1} = (1-\delta)k_t + i_t$. When $\beta(1+R) > 1$, the debt terms vanish ($b_{t+1} = 0$). The firm's action-value function reduces correspondingly to $Q^*(z_t, k_t, i_t)$, and the firm need only

[3] This problem is also computationally intractable and cannot be solved in this continuous state and action space setting.

maintain beliefs $Q \sim \mathcal{GP}(\hat{Q}_t, \hat{\Sigma}_t)$ over the three-dimensional state-action space (z_t, k_t, i_t). Proof in Appendix C.1.

2.4 Calibration & Implementation Details

The firm's Gaussian process belief over the action-value function requires specifying kernel hyperparameters, raising the question of how to discipline their calibration. This may seem worrisome at first, as calibration requires taking a stance on the scale of variation in the state-action space. In this section we provide a principled basis for this choice.

Ultimately, the kernel encodes how quickly the firm believes the value function changes as it moves from $\mathbf{x}$ to $\mathbf{x}'$ in the state-action space. This belief can be compared to the true variation in the value function, providing a measure of how appropriately the firm generalizes from past experience to new situations. If the firm believes the action-value function changes more slowly than it actually does, it over-generalizes: it treats dissimilar states as interchangeable and transfers learning too broadly. If the firm believes the action-value function changes more rapidly than it actually does, it under-generalizes: it treats similar states as distinct and fails to transfer learning across them. If the firm believes the action-value function changes at the same rate as it actually does, we consider it to be rational. We provide a formal characterization of this (Definition 1).

Definition 1 (Kernel Rationality). *An agent's GP prior $\mathcal{GP}(\hat{Q}_0, \hat{\Sigma}_0)$ is rational at the steady state $\mathbf{x}^*$ if the believed variation in Q^* equals the true variation. Define the rationality ratio along dimension j at displacement Δx as*

$$R_j(\Delta x) \equiv \frac{\operatorname{Std}\left(Q^*(\mathbf{x}^* + \Delta x\, \mathbf{e}_j) - Q^*(\mathbf{x}^*) \mid \hat{\Sigma}_0\right)}{|Q^*(\mathbf{x}^* + \Delta x\, \mathbf{e}_j) - Q^*(\mathbf{x}^*)|} \tag{19}$$

i) Locally rational at Δx: $R_j(\Delta x) = 1$ for each dimension j.

ii) Globally rational: $R_j(\Delta x) = 1$ for all $\Delta x > 0$ and all j.

When $R_j > 1$, the firm under-generalizes: it believes Q^* varies more than it truly does, treating similar states as distinct and failing to transfer learning across them. When $R_j < 1$, the firm over-generalizes: it believes Q^* varies less than it truly does, treating distinct states as equivalent and transferring learning too broadly.

Note that we distinguish between local and global rationality. The true variation in Q^* is governed by the economic primitives, whereas the kernel is a finite-parameter object that generally cannot capture the true rate of change in Q^* at every displacement simultaneously. For example, the action-value function for our firm is approximately linear in productivity z and capital k but quadratic in investment i (Fig. 2). The squared-exponential kernel, a standard choice in the GP literature and the one employed by Ilut and Valchev [2022], saturates at large displacements and therefore cannot reproduce the true variation in Q^* in each dimension (z, k, i) at every displacement Δx. A kernel calibrated to match local variation near the steady state will over-generalize at large displacements, while a kernel calibrated to match large displacements will under-generalize locally. It is therefore prudent to calibrate the kernel at a specific displacement. Three choices are natural: the local neighborhood around the steady state (gradient matching); one standard deviation of the ergodic distribution under the rational policy (the typical displacement the firm experiences); and the kernel's own characteristic scale, where the believed standard deviation equals σ_0.

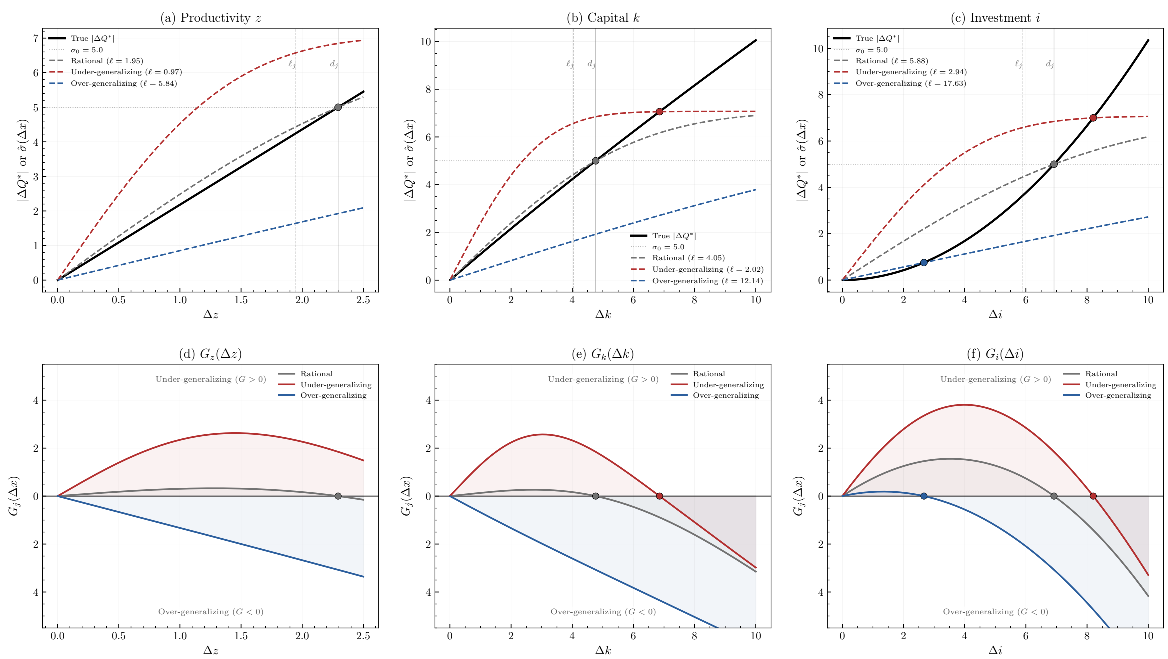

Figure 2: Kernel rationality and generalization. Top row: true variation $|\Delta Q^*|$ (black) versus believed variation $\hat{\sigma}_j(\Delta x)$ under rational (gray), under-generalizing (red), and over-generalizing (blue) calibrations. Bottom row: generalization gap $G_j(\Delta x) = \hat{\sigma}_j - |\Delta Q^*|$; positive regions indicate under-generalization, negative regions over-generalization. Dots mark crossings where the kernel is locally rational.

Fig. 2 illustrates this limitation in the firm investment problem. The squared-exponential kernel (RBF) cannot simultaneously match the true variation in Q^* at all displacements: its believed standard deviation is concave in $|\Delta x|$ while the true variation may be linear, concave, or convex depending on the dimension. At best, the believed and true variation curves cross at a single displacement, with the kernel under-generalizing ($G > 0$) on one side of the crossing and over-generalizing ($G < 0$) on the other.

$$G_j(\Delta x) \equiv \hat{\sigma}_j(\Delta x) - |Q^*(\mathbf{x}^* + \Delta x\, \mathbf{e}_j) - Q^*(\mathbf{x}^*)|\,, \tag{20}$$

Proposition 3 characterizes the crossing point $G_j(\Delta x) = 0$ in terms of the kernel length scale ℓ_j, providing a calibration target. When the prior is exactly rational along dimension j (so that $|\Delta Q_j^*| = \sigma_0$ at the crossing), the proposition yields $\ell_j = d_j/\sqrt{2\ln 2} \approx 0.849\, d_j$, explaining the gap between ℓ_j and d_j visible in the top row of Figure Fig. 2.

Proposition 3 (Rational Length Scales under the RBF Kernel)**.** *Under the RBF kernel* $\hat{\Sigma}_0(\mathbf{x}, \mathbf{x}') = \sigma_0^2 \exp\left(-\sum_j \frac{(x_j - x'_j)^2}{2\ell_j^2}\right)$, *the agent's GP prior is locally rational at* $\mathbf{x}^*$ *and displacement* d_j *if and only if, for each dimension* j*, the length scale satisfies:*

$$\ell_j = \frac{d_j}{\sqrt{-2\ln\left(1 - \frac{|\Delta Q_j^*|^2}{2\sigma_0^2}\right)}} \tag{21}$$

where $|\Delta Q_j^*| \equiv |Q^*(\mathbf{x}^* + d_j\,\mathbf{e}_j) - Q^*(\mathbf{x}^*)|$ *is the true variation in the action-value function over the chosen displacement.*

Implementation. We note that GP regression can be implemented via its kernel (dual) representation which involves inverting the $N \times N$ Gram matrix rather than operating in the potentially high-dimensional feature space.[4] In our setting the state-action space is continuous, and it is reasonable to maintain hundreds of observations in the firm's experience buffer. This representation is detailed in Appendix A.3 and is the one used in our implementation.

3 Simulations

We now turn to numerical simulations of the model. We pick a standard calibration of the investment environment: $\beta = 0.97$ stochastic discount factor, $\alpha = 0.3$ capital share, $\delta = 0.04$ depreciation rate, $R = 0.01$ gross interest rate, and $\sigma_z = 0.05$ with persistence $\rho_z = 0.9$ for the idiosyncratic productivity process (not representative of current conditions but in-line with common conditions, also not representative of any particular industry). We simulate a population of $N = 1{,}000$ firms for $T = 300$ periods, with a 2% exit rate per period. Upon exit, a new firm inherits the capital of the incumbent but starts with a blank slate of knowledge (i.e., the original prior and no experience). The behavioral distribution is determined to become ergodic by $T = 100$, so we discard these first 100 periods as burn-in. We also maintain a "shadow" rational agent to benchmark the learning agents' behavior against. Note that because each agent faces identical environments, they ideally should converge to the same optimal policy, so any cross-sectional dispersion in behavior is solely due to differences in learning and reasoning dynamics generated by their idiosyncratic experiences.

3.1 Experience-Only Simulations

We first simulate using experience-only learning: the firms only update their beliefs based on the observed rewards and transitions from their actions, without any reasoning-based deliberation. In this way we can understand how the learning process alone shapes firm behavior and heterogeneity. Second we can study the proper calibration of the reasoning mechanism.

[4] This is more computationally efficient when the number of observations N is small relative to the feature-space dimensionality.

The current hypothesis is as follows: because the belief sequence is a martingale under certainty equivalence, reasoning does not shift the conditional mean of Q^*. Its sole effect is to compress the posterior variance, which in turn tightens the entropy floor governing exploration. This compression can be beneficial or detrimental depending on where the firm's beliefs stand relative to the truth. When the firm's current policy is close to optimal, reasoning usefully stabilizes beliefs and prevents unnecessary experimentation from pulling the firm away. When the firm has already deviated from the optimal policy, however, the same variance reduction locks in the deviation: the tighter posterior forecloses the exploration that would be needed to discover and correct the error.

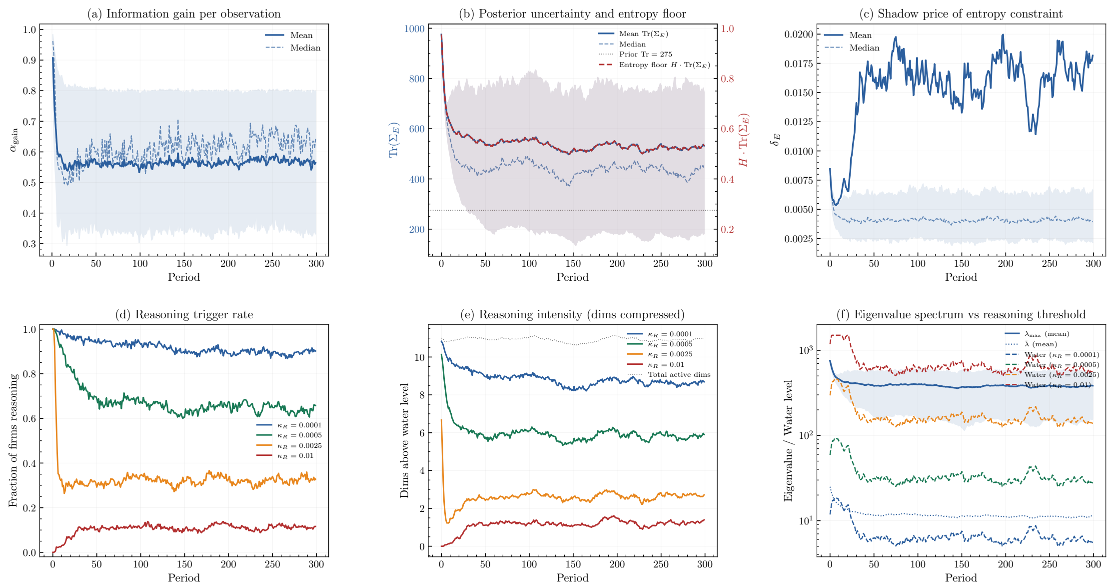

Figure 3: Learning dynamics for population of firms under experience-only learning (N = 1,000 firms, T = 300, burn-in $T > 100$). Panel (a): information gain α_t from experience updates; Panel (b): posterior uncertainty Σ_t plotted against the entropy floor constraint $H \cdot \mathrm{Tr}(\Sigma_E)$; Panel (c): Shadow price δ_E of information; Panel (d): fraction of firms triggering reasoning mechanism plotted against multiple costs κ. Panel (e): intensity of reasoning measured by dimensions of belief space compressed by reasoning, plotted against κ. Panel (f): eigenvalue spectrum vs. reasoning threshold ($\kappa/(H \cdot \delta_E)$) across firms.

Fig. 3 plots the learning dynamics across the population of firms under experience-only learning. The first thing to note from Panels (a) and (b) is that the information gain from experience updates stabilizes above zero, and posterior uncertainty remains above the entropy floor. The firms suffer from a curse of dimensionality in learning. Given they are short-lived with an average lifespan of 50 periods (due to the 2% exit rate), they accumulate roughly 50 observations of the action-value function over a three-dimensional input space (z, k, i). Uniform coverage of a d-dimensional space requires $n^{1/d}$ observations per dimension, so 50 observations in three dimensions yields approximately $50^{1/3} \approx 3.7$ effective observations per axis. Sparse coverage explains why the information gain α_t stabilizes above zero and the posterior trace remains elevated.

The second row of Fig. 3 quantifies how reasoning would interact with the experience-based learning dynamics. At moderate costs ($\kappa_R = 0.0025$), roughly 40% of firm-periods would trigger reasoning, compressing on average 3 eigenvalue dimensions per episode. Panel (f) confirms that the water level $\kappa_R/(H \cdot \delta_E)$ sits within the eigenvalue spectrum rather than above or below it, so reasoning selectively compresses the dominant directions of uncertainty while leaving already-resolved dimensions untouched. As reasoning becomes cheaper, the water level drops and more dimensions are compressed, accelerating belief convergence but also increasing the risk of premature lock-in to policies shaped by early, potentially unrepresentative experience.

	Investment rate i/k			Output ratio y/k			Difference
H	Mean	Std	$\overline{\sigma}_\times$	Mean	Std	Δ	$\Delta\, d/k$ (%)
5×10^{-5}	0.0409	0.0444	0.0440	0.2291	0.0887	0.0209	−11.3
1×10^{-4}	0.0411	0.0499	0.0496	0.2364	0.1004	0.0282	−15.3
2.5×10^{-4}	0.0414	0.0531	0.0529	0.2385	0.1060	0.0303	−16.3
5×10^{-4}	0.0413	0.0530	0.0527	0.2488	0.1101	0.0406	−22.6
7.5×10^{-4}	0.0416	0.0540	0.0537	0.2419	0.1098	0.0337	−18.2
1×10^{-3}	0.0415	0.0535	0.0532	0.2399	0.1096	0.0317	−17.0
2.5×10^{-3}	0.0424	0.0607	0.0602	0.2309	0.0979	0.0227	−10.6
Rational	0.0401	0.0124	—	0.2082	0.0176	—	0.0

Table 1: Dispersion statistics under experience-only learning by entropy parameter H. $N =$ 1,000 firms, $T = 300$ periods, ergodic sample $t \geq 100$. Std(i/k) is the total standard deviation of the investment rate (time and cross-section pooled); $\overline{\sigma}_\times(i/k)$ is the average cross-sectional standard deviation at each period, isolating pure policy dispersion from time-series variation. $\Delta d/k$ reports the percentage deviation of mean d/k from the rational benchmark.

Table 1 quantifies how the entropy parameter H shapes the ergodic distribution of firm behavior under experience-only learning. Even at the lowest H, the cross-sectional standard deviation of investment rates $\overline{\sigma}_\times(i/k)$ is roughly four times the rational benchmark, reflecting persistent heterogeneity in learned policies that would not arise under rational expectations. As H increases, policy dispersion rises monotonically: higher entropy floors force broader exploration, which exposes firms to more diverse (and noisier) regions of the state space, producing greater cross-sectional variation in converged beliefs. The mean investment rate $\overline{i/k}$ is largely insensitive to H, remaining within one percentage point of the rational value across all calibrations. This dispersion has a compositional consequence for the reported ratios. Because some experience-learning firms under-invest and shrink, the average capital stock falls below the rational steady state. Under concave production ($\alpha < 1$), output per unit capital $y/k = zk^{\alpha-1}$ is decreasing in k, so smaller firms mechanically exhibit higher y/k. The same logic extends to d/k: since dividends inherit the concavity of the production function through the flow payoff, which is why we higher dividend yield d/k and negative deviation from the rational benchmark.

3.2 Experience and Reasoning Simulations

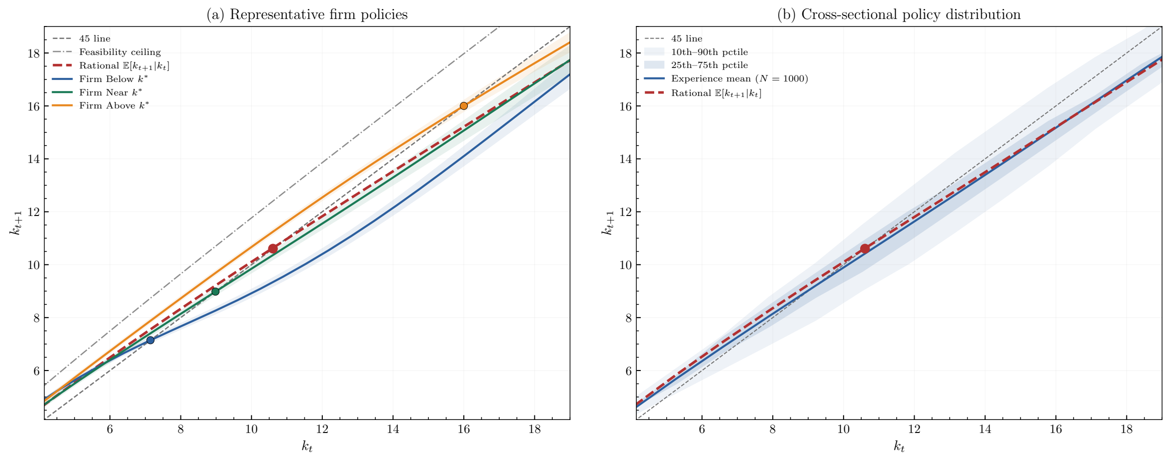

Figure 4: Expected policy functions ($N = 1{,}000$ firms, $T = 300$ periods, 2% firm exit rate). Panel (a): expected policy $\mathbb{E}[k_{t+1} \mid k_t]$ for three representative firms selected from the cross-sectional distribution, with shaded bands showing ± 1 standard deviation of the softmax policy. Panel (b): cross-sectional policy envelope, showing the mean $\mathbb{E}[k_{t+1} \mid k_t]$ across all firms (solid blue), with the 25th–75th percentile band (dark shading) and 10th–90th percentile band (light shading). The rational benchmark policy is shown in red (dotted). The dash-dotted line indicates the dividend feasibility constraint: the maximum k_{t+1} such that $d_t \geq 0$ at $z = 1$.

Fig. 4 shows the results of the simulations. Here we can see that firm heterogeneity endogenously emerges. The population of firms develops a distribution of policies, with some firms being capital intensive and others being more conservative. The cross-sectional policy envelope shows that the mean policy is close to the rational benchmark, but there is significant dispersion around it.

The previous figure only shows the resulting distribution of policies at end of simulation, but to show that the distribution is truly ergodic, we can also look at the distribution of firm-level ratios across time and firms. Fig. 5 does just this, showing the distribution of the investment rate i_t/k_t, output–capital ratio y_t/k_t, and dividend yield d_t/k_t across all firms and periods. Together, these figures show that the model can rationalize investment dispersion (challenge #3) and lumpy investment dynamics (challenge #4) as emergent properties.

One notable feature is the multi-modality of the ergodic distribution of i_t/k_t in panel (a). This might arise from a shape mismatch between the kernel and the true action-value function: the squared-exponential kernel is concave in displacement (its covariance saturates at large Δi), whereas Q^* is convex in investment due to quadratic adjustment costs (its curvature increases with $|\Delta i|$). As a result, the agent under-generalizes at small deviations from its current investment level (believing Q^* varies more than it does locally) while over-generalizing at large deviations (believing Q^* varies less than it does in the tails).

To test for investment dynamics, specifically the presence of sensitivity to q or cash flow,

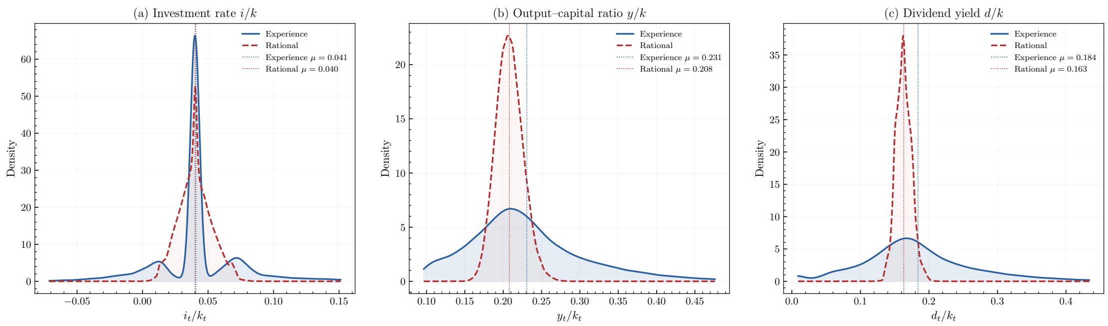

Figure 5: Ergodic distributions of firm-level ratios under experience-only learning ($N = 1{,}000$ firms, $T = 300$, burn-in $T > 100$). Panel (a): investment rate i_t/k_t, Panel (b): output–capital ratio y_t/k_t. Panel (c): dividend yield d_t/k_t.

we run the Fazzari et al. [1988] regression of i_t/k_t on q_t and CF_t/k_t. The rational agent, as expected, has a strong sensitivity to q and no sensitivity to cash flow after. The learning agent, however, has a much weaker sensitivity to q rationalizing the empirical finding that q is a weak predictor of investment (challenge #1). However, the learning agent does not have a significant sensitivity to cash flow after controlling for q, which is not consistent with the empirical finding that cash flow has significant predictive power for investment (challenge #2). So our model cannot rationalize the cash-flow sensitivity empirical regularity. These results are shown in Fig. 6.

Finally, we examine the distribution of capital across firms. Figure Fig. 7 plots the Lorenz curve of time-averaged capital holdings across firms alongside the cross-sectional Gini coefficient at each period. The economy populated by learning and reasoning agents exhibits greater capital inequality than the rational reference benchmark, in which identical shadow agents follow the optimal policy under the same shock sequences. This heterogeneity arises endogenously from the learning mechanism: firms that receive favorable early shock sequences develop locally confident (but idiosyncratic) beliefs about optimal investment through both experience and reasoning, leading to persistent dispersion in capital accumulation that would not emerge under rational expectations.

We now characterize how the reasoning mechanism shapes the learning process.

Proposition 4 (Cognitive Friction and the Exploration Floor)**.** *Let π_t denote the optimal policy after the experience update and reasoning cycle, with differential entropy $\mathcal{H}(\pi_t) = -\int_{\mathbb{R}} \pi_t(c) \ln \pi_t(c)\, dc$.*

i) For any cognitive cost $\kappa > 0$, there exists an absolute lower bound on policy entropy $\underline{\mathcal{H}}(\kappa) \equiv \inf_{\pi_t} \mathcal{H}(\pi_t)$ such that:

$$\mathcal{H}(\pi_t) \geq \underline{\mathcal{H}}(\kappa) > 0 \quad \forall t. \tag{22}$$

ii) For all κ below the shutdown threshold $\bar{\kappa} = H \cdot \delta_E \cdot \sup_i \lambda_i^E$, the exploration floor is

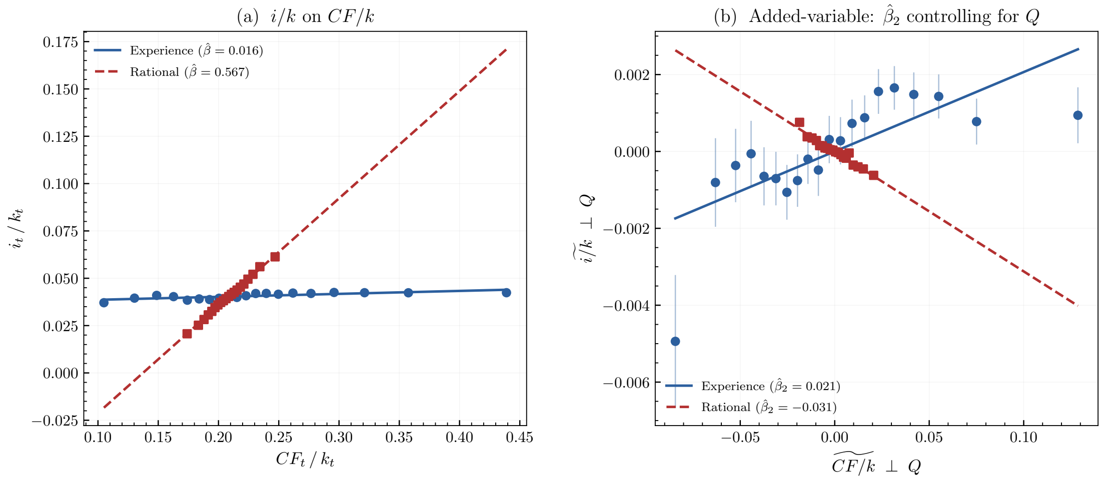

Figure 6: Investment regression (N = 1,000 firms, T = 300, ergodic sample $t > 100$). Panel (a): naive binned scatter of i_t/k_t on CF_t/k_t without controls. Panel (b): added-variable plot after partialling out Q_t, isolating the coefficient $\hat{\beta}_2$ from the specification $i_t/k_t = a + \beta_1 Q_t + \beta_2(CF_t/k_t) + \varepsilon_t$. For the rational agent, Q is a sufficient statistic for investment ($\hat{\beta}_1 = 0.42$, $R^2 = 0.99$) and cash flow has no residual explanatory power ($\hat{\beta}_2 \approx 0$).

strictly increasing in the cost of reasoning:

$$\frac{\partial \underline{\mathcal{H}}}{\partial \kappa} = \frac{1}{\delta_E} \sum_{i=1}^{\infty} \mathbb{1}_{\left\{\lambda_i^E > \frac{\kappa}{H \cdot \delta_E}\right\}} > 0. \tag{23}$$

Consequently, for any $\bar{\kappa} > \kappa_2 > \kappa_1 > 0$,

$$\underline{\mathcal{H}}(\kappa_2) > \underline{\mathcal{H}}(\kappa_1). \tag{24}$$

Proof provided in Appendix C.2.

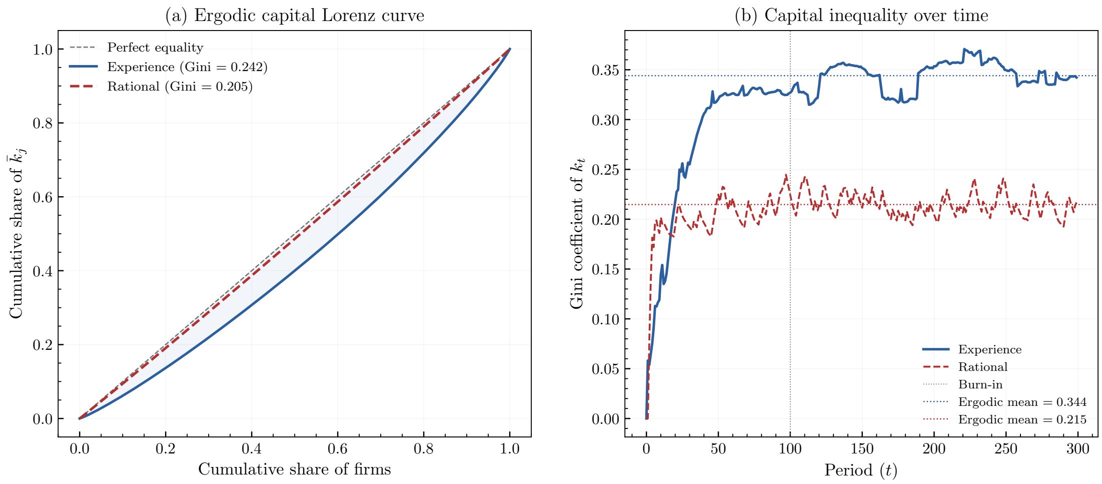

Figure 7: Capital inequality under experience-only learning ($N = 1{,}000$ firms, $T = 300$, burn-in $T > 100$). Panel (a): Lorenz curve of time-averaged capital $\bar{k}_j = T_{\text{erg}}^{-1} \sum_{t>t_0} k_{jt}$ across firms, with Gini coefficients reported. Panel (b): cross-sectional Gini coefficient of k_t at each period.

A Derivations

A.1 Derivation of the Canonical Investment Model Solution

This section derives the optimal investment condition for the firm described in Section Section 2. The firm chooses investment $i_{j,t}$ and next-period debt $b_{j,t+1}$ to maximize discounted dividends subject to the capital accumulation equation and the collateral borrowing constraint. Let $q_{j,t}$ denote the multiplier on the capital accumulation equation and $\mu_{j,t}$ the multiplier on the collateral constraint. The Lagrangian is

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big[f(z_{j,t}, k_{j,t}) - i_{j,t} - \frac{\phi}{2} \left(\frac{i_{j,t}}{k_{j,t}} \right)^2 k_{j,t} + b_{j,t+1} - (1+r) b_{j,t} \tag{25}$$

$$+ q_{j,t} \big((1-\delta) k_{j,t} + i_{j,t} - k_{j,t+1} \big) \tag{26}$$

$$+ \mu_{j,t} \big(\theta k_{j,t+1} - b_{j,t+1} \big) \Big]. \tag{27}$$

The first-order condition with respect to investment is

$$-1 - \phi \frac{i_{j,t}}{k_{j,t}} + q_{j,t} = 0. \tag{28}$$

Rearranging gives the Tobin's q investment condition

$$q_{j,t} = 1 + \phi \frac{i_{j,t}}{k_{j,t}}. \tag{29}$$

Solving for the investment rate,

$$\frac{i_{j,t}}{k_{j,t}} = \frac{1}{\phi}(q_{j,t} - 1). \tag{30}$$

Under quadratic adjustment costs, the investment-capital ratio is linear in marginal q. The first-order condition with respect to next-period debt is

$$1 - \beta(1 + r) - \mu_{j,t} = 0. \tag{31}$$

The first-order condition with respect to next-period capital is

$$-q_{j,t} + \beta\mathbb{E}_t\Big[f_k(z_{j,t+1}, k_{j,t+1}) + (1 - \delta)q_{j,t+1} + \theta\mu_{j,t+1}\Big] = 0. \tag{32}$$

Rearranging yields the Euler equation for marginal q,

$$q_{j,t} = \beta\mathbb{E}_t\Big[f_k(z_{j,t+1}, k_{j,t+1}) + (1 - \delta)q_{j,t+1} + \theta\mu_{j,t+1}\Big]. \tag{33}$$

The borrowing constraint satisfies the complementary slackness conditions

$$b_{j,t+1} \leq \theta k_{j,t+1} \tag{34}$$

$$\mu_{j,t} \geq 0 \tag{35}$$

$$\mu_{j,t}\big(\theta k_{j,t+1} - b_{j,t+1}\big) = 0. \tag{36}$$

When the constraint is slack ($\mu_{j,t} = 0$), the model reduces to the standard Tobin's q investment condition. When the constraint binds ($\mu_{j,t} > 0$), capital carries an additional value because it relaxes the borrowing limit.

A.2 Derivation of Optimal Action Policy Proposition 7

The agent maximizes expected payoff subject to information and probability constraints:

$$\begin{aligned}
&\max_{\pi_t(\cdot|s_t)} \sum_c \hat{Q}_t(c, s_t)\pi_t(c|s_t) && \text{(Objective Function)}\\
\text{s.t.}\quad & -\sum_c \pi_t(c|s_t)\ln \pi_t(c|s_t) \geq \kappa \sum_c \sigma_t(c|s_t) && \text{(Information Constraint)}\\
&\sum_c \pi_t(c|s_t) = 1 && \text{(Probability Constraint)}
\end{aligned}$$

The Lagrangian and subsequent First-Order Conditions (FOC) are:

$$
\begin{aligned}
\mathcal{L} &= \sum_c \hat{Q}_t(c, s_t)\pi_t(c|s_t) + \delta_t \left[-\sum_c \pi_t \ln \pi_t - \kappa \sum \sigma \right] + \mu \left[\sum_c \pi_t - 1 \right] && \text{(Lagrangian)} \\
\frac{\partial \mathcal{L}}{\partial \pi_t} &= \hat{Q}_t(c, s_t) - \delta_t \ln \pi_t(c|s_t) - \delta_t + \mu = 0 && \text{(FOC w.r.t. } \pi_t) \\
&\implies \ln \pi_t(c|s_t) = \frac{\mu - \delta_t}{\delta_t} + \frac{\hat{Q}_t(c, s_t)}{\delta_t} && \text{(Solve for } \ln \pi_t)
\end{aligned}
$$

Solving for the optimal policy $\pi_t(c|s_t)$:

$$
\begin{aligned}
\pi_t(c|s_t) &= \exp\left(\frac{\mu - \delta_t}{\delta_t}\right) \exp\left(\frac{\hat{Q}_t(c, s_t)}{\delta_t}\right) && \text{(Exponentiate)} \\
&= A \cdot \exp\left(\frac{\hat{Q}_t(c, s_t)}{\delta_t}\right) && \text{(Let } A = e^{\frac{\mu-\delta_t}{\delta_t}}) \\
&= \frac{\exp\left(\frac{\hat{Q}_t(c, s_t)}{\delta_t}\right)}{\sum_{c'} \exp\left(\frac{\hat{Q}_t(c', s_t)}{\delta_t}\right)} && \text{(Normalize via } \sum \pi_t = 1)
\end{aligned}
$$

In the limit where the shadow price of attention $\delta_t \to 0$:

$$
\lim_{\delta_t \to 0} \pi_t(c|s_t) = \begin{cases} 1 & \text{if } c = \arg\max_{c'} \hat{Q}_t(c', s_t) \\ 0 & \text{otherwise} \end{cases}
$$

A.3 Gaussian Process Regression & Experience Learning

The primal update in Definition 6 produces three objects: $\hat{Q}_t^E$, $\hat{\Sigma}_t^E$, and α_t^E. Each of these objects is a function over the continuous domain $\mathcal{A} \times \mathcal{S}$. To implement the primal form directly, one would need to evaluate and store $\hat{Q}_t^E(\mathbf{a}, \mathbf{s})$ at every point in the action-state space, which is uncountably infinite. In the code, we instead immplement a GP regression which sidesteps this issue by storing not the updated functions themselves, but the finite collection of data from which those functions can be reconstructed at any query point on demand.

From $t = 1$. In the first period, the agent has some prior $Q_0^* \sim GP(\mu, k)$ they start in state $\mathbf{s}_0$ and land in state $\mathbf{s}_1$ after taking actions $\mathbf{a}_0$. They observe a noisy signal, their utility from that action in that state, which is a noisy observation of the linear functional $Q^*(\mathbf{a}_0, \mathbf{s}_0) - Q^*(\mathbf{a}_1, \mathbf{s}_1) = \eta_1^E - \varepsilon_1$. The noise comes from the stochastic transition. As a result, δ_1, the temporal difference error, is a noisy signal of the linear functional $Q^*(\mathbf{a}_0, \mathbf{s}_0) - \beta Q^*(\mathbf{a}_1, \mathbf{s}_1)$.

By the linearity of covariance.

$$\mathrm{Cov}(\delta_1, Q^*(\mathbf{a},\mathbf{s})) = k((\mathbf{a}_0,\mathbf{s}_0),(\mathbf{a},\mathbf{s})) - \beta k((\mathbf{a}_1,\mathbf{s}_1),(\mathbf{a},\mathbf{s})) \tag{37}$$

$$\mathrm{Var}(\delta_1) = k((\mathbf{a}_0,\mathbf{s}_0),(\mathbf{a}_0,\mathbf{s}_0)) + \beta^2 k((\mathbf{a}_1,\mathbf{s}_1),(\mathbf{a}_1,\mathbf{s}_1)) - 2\beta k((\mathbf{a}_0,\mathbf{s}_0),(\mathbf{a}_1,\mathbf{s}_1)) \tag{38}$$

Denote $w \equiv \mathrm{Var}(\delta_1)$, the denominator above. The posterior mean at any query point $(\mathbf{a},\mathbf{s})$ then follows directly from Definition 6:

$$\hat{Q}_1^E(\mathbf{a},\mathbf{s}) = \mu(\mathbf{a},\mathbf{s}) + \alpha_1^E(\mathbf{a},\mathbf{s}) \cdot \delta_1 = \mu(\mathbf{a},\mathbf{s}) + \frac{k((\mathbf{a}_0,\mathbf{s}_0),(\mathbf{a},\mathbf{s})) - \beta\, k((\mathbf{a}_1,\mathbf{s}_1),(\mathbf{a},\mathbf{s}))}{w} \cdot \delta_1 \tag{39}$$

where $\delta_1 = \eta_1^E + \beta\,\mu(\mathbf{a}_1,\mathbf{s}_1) - \mu(\mathbf{a}_0,\mathbf{s}_0)$ is the temporal difference error evaluated against the prior mean.

After N observations, the dual form maintains four objects. First, two $N \times d$ matrices of support points: the *decision points* $\mathbf{X}^{\mathrm{dec}} = \{(\mathbf{a}_0,\mathbf{s}_0),\ldots,(\mathbf{a}_{N-1},\mathbf{s}_{N-1})\}$, where actions were taken, and the *outcome points* $\mathbf{X}^{\mathrm{out}} = \{(\mathbf{a}_1,\mathbf{s}_1),\ldots,(\mathbf{a}_N,\mathbf{s}_N)\}$, where the agent landed. Second, the N-vector of observed utilities $\mathbf{Y} = (\eta_1^E,\ldots,\eta_N^E)^\top$. Third, an $N \times N$ matrix C^{-1}, the inverse of the Gram matrix C.

Gram matrix. The Gram matrix C is the $N \times N$ matrix whose (i,j) entry is the prior covariance between the i-th and j-th TD functionals:

$$C_{ij} = \mathrm{Cov}(\delta_i,\delta_j) = k(\mathbf{x}_i^{\mathrm{dec}},\mathbf{x}_j^{\mathrm{dec}}) - \beta\, k(\mathbf{x}_i^{\mathrm{dec}},\mathbf{x}_j^{\mathrm{out}}) - \beta\, k(\mathbf{x}_i^{\mathrm{out}},\mathbf{x}_j^{\mathrm{dec}}) + \beta^2\, k(\mathbf{x}_i^{\mathrm{out}},\mathbf{x}_j^{\mathrm{out}}) \tag{40}$$

The four-term expansion follows from the bilinearity of covariance applied to the linear functional $\delta_i = Q^*(\mathbf{x}_i^{\mathrm{dec}}) - \beta\, Q^*(\mathbf{x}_i^{\mathrm{out}}) + \varepsilon_i$. Expanding $\mathrm{Cov}(Q^*(\mathbf{x}_i^{\mathrm{dec}}) - \beta\, Q^*(\mathbf{x}_i^{\mathrm{out}}),\ Q^*(\mathbf{x}_j^{\mathrm{dec}}) - \beta\, Q^*(\mathbf{x}_j^{\mathrm{out}}))$ and using $\mathrm{Cov}(Q^*(\mathbf{x}), Q^*(\mathbf{x}')) = k(\mathbf{x},\mathbf{x}')$ yields the four kernel evaluations above. Intuitively, C_{ij} measures how much the i-th and j-th temporal difference observations tell us about one another under the prior.

Prior TD means. Define the N-vector of prior TD means $\mathbf{M}$, whose i-th entry is the prior expectation of the i-th TD functional:

$$M_i = \mu(\mathbf{a}_{i-1},\mathbf{s}_{i-1}) - \beta\,\mu(\mathbf{a}_i,\mathbf{s}_i) \tag{41}$$

The residual $\mathbf{Y} - \mathbf{M}$ is then the N-vector whose i-th entry is $\eta_i^E - M_i = \eta_i^E - \mu(\mathbf{a}_{i-1},\mathbf{s}_{i-1}) + \beta\,\mu(\mathbf{a}_i,\mathbf{s}_i) = \delta_i$, the temporal difference error of the i-th observation evaluated against the *original prior* μ. In the primal (Kalman) form of Definition 6, the TD error at step t is evaluated against the current posterior mean $\hat{Q}_{t-1}^E$, which has already absorbed all previous observations. In the dual form, every residual is evaluated against the same unchanging prior μ. The two forms nonetheless yield identical posteriors because the inverse Gram matrix C^{-1} compensates for this difference.

Decorrelation. To reconstruct the posterior at any query point $(\mathbf{a}, \mathbf{s})$, we compute the N-vector $k_*(\mathbf{a}, \mathbf{s})$ whose i-th entry is the cross-covariance between the i-th TD functional and the latent Q^* at the query point:

$$[k_*(\mathbf{a}, \mathbf{s})]_i = k((\mathbf{a}_{i-1}, \mathbf{s}_{i-1}), (\mathbf{a}, \mathbf{s})) - \beta\, k((\mathbf{a}_i, \mathbf{s}_i), (\mathbf{a}, \mathbf{s})) \tag{42}$$

The posterior mean and variance are then:

$$\hat{Q}_N^E(\mathbf{a}, \mathbf{s}) = \mu(\mathbf{a}, \mathbf{s}) + \mathbf{k}_*(\mathbf{a}, \mathbf{s})^\top C^{-1} (\mathbf{Y} - \mathbf{M}) \tag{43}$$

$$\hat{\Sigma}_N^E((\mathbf{a}, \mathbf{s}), (\mathbf{a}, \mathbf{s})) = \sigma_0^2 - k_*(\mathbf{a}, \mathbf{s})^\top C^{-1} k_*(\mathbf{a}, \mathbf{s}) \tag{44}$$

$C^{-1}(\mathbf{Y} - \mathbf{M})$ determines how much each past observation revises the belief at a query point. C^{-1} performs decorrelation simultaneously across all observations. The inverse Gram matrix thus converts the naive prior-based residuals $\mathbf{Y} - \mathbf{M}$ into the sequentially conditioned innovations that the primal Kalman recursion in Definition 6 would produce. Dual form then allows us to reconstruct the posterior at any query point without ever storing the posterior mean and variance functions themselves.

Worked example $t = 1$. The agent begins with prior $Q^* \sim \mathcal{GP}(\mu, k)$, takes action $\mathbf{a}_0$ in state $\mathbf{s}_0$, receives utility $\eta_1^E = u(\mathbf{a}_0, \mathbf{s}_0)$, and transitions to state $\mathbf{s}_1$. Define $\mathbf{x}_1^{\text{dec}} = (\mathbf{a}_0, \mathbf{s}_0)$ and $\mathbf{x}_1^{\text{out}} = (\tilde{\mathbf{a}}_1, \mathbf{s}_1)$, where $\tilde{\mathbf{a}}_1 = \arg\max_{\mathbf{a}} \mu(\mathbf{a}, \mathbf{s}_1)$. The dual form stores these support points, the scalar $\mathbf{Y} = (\eta_1^E)$, and the 1×1 inverse Gram matrix $C^{-1} = (C_{11})^{-1}$, where

$$C_{11} = k(\mathbf{x}_1^{\text{dec}}, \mathbf{x}_1^{\text{dec}}) - 2\beta\, k(\mathbf{x}_1^{\text{dec}}, \mathbf{x}_1^{\text{out}}) + \beta^2\, k(\mathbf{x}_1^{\text{out}}, \mathbf{x}_1^{\text{out}}) = \mathrm{Var}(\delta_1) \tag{45}$$

is the prior variance of the first TD functional. The prior TD mean is $M_1 = \mu(\mathbf{a}_0, \mathbf{s}_0) - \beta\, \mu(\tilde{\mathbf{a}}_1, \mathbf{s}_1)$, so the residual $Y_1 - M_1 = \delta_1$. The weight is $\alpha_1 = \delta_1 / C_{11}$. At any query point $(\mathbf{a}, \mathbf{s})$, define $k_{*1} = k(\mathbf{x}_1^{\text{dec}}, (\mathbf{a}, \mathbf{s})) - \beta\, k(\mathbf{x}_1^{\text{out}}, (\mathbf{a}, \mathbf{s}))$. Then:

$$\hat{Q}_1^E(\mathbf{a}, \mathbf{s}) = \mu(\mathbf{a}, \mathbf{s}) + \frac{k_{*1}}{C_{11}} \cdot \delta_1 = \mu(\mathbf{a}, \mathbf{s}) + \alpha_1^E(\mathbf{a}, \mathbf{s}) \cdot \delta_1 \tag{46}$$

The ratio k_{*1}/C_{11} is precisely the Kalman gain at step 1. The posterior variance is $\hat{\Sigma}_1^E = \sigma_0^2 - k_{*1}^2 / C_{11}$.

Worked example $t = 2$. A second transition yields η_2^E, with new support points $\mathbf{x}_2^{\text{dec}} = (\mathbf{a}_1, \mathbf{s}_1)$ and $\mathbf{x}_2^{\text{out}} = (\tilde{\mathbf{a}}_2, \mathbf{s}_2)$. The Gram matrix expands to 2×2. The new diagonal entry $C_{22} = \mathrm{Var}(\delta_2)$ has the same structure as C_{11}. The off-diagonal $C_{12} = \mathrm{Cov}(\delta_1, \delta_2)$ measures how correlated the two TD observations are (large when the observations are in nearby regions). Schur complement allows us to efficiently update the inverse Gram matrix without inverting the full 2×2 matrix.

$$S = C_{22} - \frac{C_{12}^2}{C_{11}} = \mathrm{Var}(\delta_2 \mid \delta_1) \tag{47}$$

The updated inverse is:

$$C_{\text{new}}^{-1} = \begin{pmatrix} \frac{1}{C_{11}} + \frac{q^2}{S} & -\frac{q}{S} \\ -\frac{q}{S} & \frac{1}{S} \end{pmatrix}, \qquad q = \frac{C_{12}}{C_{11}} \tag{48}$$

The weight vector $\boldsymbol{\alpha} = C^{-1}(\mathbf{Y} - \mathbf{M})$ now has two entries. The second is:

$$\alpha_2 = \frac{\delta_2^{\text{prior}} - q \cdot \delta_1^{\text{prior}}}{S} \tag{49}$$

where $\delta_i^{\text{prior}} = \eta_i^E - M_i$ is the i-th TD error evaluated against the original prior μ. The numerator subtracts from the raw residual of observation 2 the component linearly predicted by observation 1 via the regression coefficient $q = C_{12}/C_{11}$. This is the incremental surprise, or the Kalman innovation at step 2, which the primal form obtains by measuring δ_2 against the updated posterior $\hat{Q}_1^E$ rather than μ. The posterior at any query point is:

$$\hat{Q}_2^E(\mathbf{a}, \mathbf{s}) = \mu(\mathbf{a}, \mathbf{s}) + k_{*1}\,\alpha_1 + k_{*2}\,\alpha_2 \tag{50}$$

The two forms yield identical posteriors: the primal form updates beliefs sequentially so that each innovation is small; the dual form measures all residuals against the prior and lets C^{-1} decorrelate them. The pattern generalizes to N observations.

B Experience and Reasoning Agent in Consumption Setting

Conside an agent with an expected utility function facing a consumption problem over a sequence of periods. At each period t, the agent chooses consumption c_t, and perfectly observes state variable s_t. The utility function is a function of c_t and s_t and therefore the agent perfectly observes their payoff $u(c_t, s_t)$ at each period. The agent discounts future utility by a factor $\beta \in (0, 1)$.

$$V^*(s_t) = \max_{c_t \in B(s_t)} \{u(c_t, s_t) + \beta \mathbb{E}\left[V^*(s_{t+1})\right]\} \tag{51}$$

This is the Bellman equation for the agent's state value function $V^*(s_t)$, where $B(s_t)$ is the agent's budget set at state s_t. This is directly linked to the agent's action value function $Q^*(s_t, c_t)$ and optimal policy function as follows.[5].

[5]Here we are using $*$ to indicate the state and action values functions under the optimal policy, there is no difference in the formation of the equations under different policies. Notationally, it would be indicated as V^π and Q^π for a policy π.

$$
\begin{aligned}
\pi^*(s) &= \arg\max_{c \in B(s)} Q^*(s,c), \\
V^*(s) &= \max_{c \in B(s)} Q^*(s,c) = Q^*(s, \pi^*(s)), \\
Q^*(s,c) &= u(c,s) + \beta \mathbb{E}[V^*(s') \mid s, c], \\
V^*(s) &= u(\pi^*(s), s) + \beta \mathbb{E}[V^*(s') \mid s, \pi^*(s)].
\end{aligned}
$$

There is a stochastic Markov transition function $\mathcal{F}(c_t, s_t, \epsilon_{t+1})$ where $\epsilon_{t+1} \sim \mathcal{G}(\epsilon_{t+1})$ is an i.i.d. shock that perturbs the transition from s_t to s_{t+1}.

B.1 Agent's Beliefs

The agent believes there to be some unknown optimal policy π^* that produces optimal state action value function $Q^*(s_t, c_t)$. As the agent has not learned this policy yet, but they do form a prior over the set of possible action value functions $\mathcal{Q}$, which are modeled as Gaussian Processes (GPs).

$$
Q^*(c,s) \sim GP\left(\hat{Q}_0(c,s), \hat{\Sigma_0}((s,c),(s',c'))\right) \tag{52}
$$

The mean function is $\hat{Q}_0(c,s)$ and the covariance function is $\hat{\Sigma_0}((s,c),(s',c'))$. The use of GPs becomes useful in that they allow the agent to have correlated beliefs over similar state-action pairs, allowing for generalization across the space $\mathcal{Q}$. As new observations are made, the agent can update their beliefs using Bayesian updating, which results in a new GP posterior over $\mathcal{Q}$.

B.2 Learning Process

Given the agent's prior is conveniently expressed as a gaussian process over action value functions, the agent performs Bayesian updating as new observations are made. The agent updates her beliefs over $\mathcal{Q}$ through two mechanisms: experience and reasoning.[6] The experience mechanism involves updating beliefs through direct observation of realized outcomes (s_t, c_t, u_t, s_{t+1}) from actions taken. The reasoning mechanism involves updating beliefs through mental simulation—evaluating counterfactual actions using the agent's model of transitions $\mathcal{F}$ and payoffs u without taking those actions.

Definition 2 (The Experience-Reasoning Learning Cycle)**.** *At each period t, the agent immediately receives signal $u(c_{t-1}, s_{t-1})$ from the previous period's action. She then first performs*

[6] Given the agent is endowed with a belief over the action value function Q^*, they can derive implied beliefs over the optimal policy π^* and state value function V^* using the relationships stated in Proposition **??**. To compute the agent's prior belief over policies π, we need to calculate the distribution of the arg max of a GP. However, this is not analytically tractable: if $f \sim GP(\mu, k)$, then $\arg\max_x f(x)$ is a random element of the input space whose distribution depends on the entire sample path structure and has no closed form.

an experience-based update to form intermediate beliefs $\hat{Q}_t^E, \hat{\Sigma}_t^E$. Which has a corresponding optimal policy π_t^E and state value function V_t^E. Next, she performs a reasoning-based update using mental simulation under the new policy π_t^E to form reasoned beliefs $\hat{Q}_t^R, \hat{\Sigma}_t^R$, with corresponding π_t^R and V_t^R. Finally, she chooses action $c_t = \pi_t^R(s_t)$ under the reasoned policy.

B.2.1 Experience-Based Learning

The agent takes a consumption action c_{t-1} in state s_{t-1}, receives payoff $u_{t-1} = u(c_{t-1}, s_{t-1})$, and transitions to state s_t, based on the Markov transition function $\mathcal{F}(c_{t-1}, s_{t-1}, \epsilon_t)$. They can now form a temporal difference (TD) approximation. By rearranging the Bellman equation we can write the realized utility as a function of the action value function.

$$\underbrace{u(c_{t-1}, s_{t-1})}_{\text{observed}} = \underbrace{Q^*(c_{t-1}, s_{t-1})}_{\text{unknown, GP prior}} - \beta \underbrace{\mathbb{E}_{t-1}[Q^*(s_t, \pi^*(s_t))]}_{\text{requires integration over } \mathcal{F}} \tag{53}$$

Since the agent does not know Q^* or the transition function $\mathcal{F}$ *ex ante*, they cannot compute the Bellman expectation analytically. In the experience-based learning phase, the agent replaces the theoretical expectation over all possible future states with the single realized transition to s_t. Rather than performing a simulation of potential outcomes, the agent learns directly from the empirical evidence of the outcome. By observing the arrived state s_t and identifying their current best guess of the optimal action, $\tilde{c}_t = \arg\max_c \hat{Q}_{t-1}(c, s_t)$, the agent constructs a Temporal Difference (TD) signal. This signal serves as a sample-based realization of the Bellman equation, which the agent then uses to perform a Bayesian update on the GP prior. This is equivalent to *model-free* learning in RL literature.

Definition 3 (Experience-Based Learning Update)**.** *Upon arriving at period t, the agent observes the realized utility $\eta_t^E = u(c_{t-1}, s_{t-1})$ and the transition to state s_t. Given the prior beliefs $(\hat{Q}_{t-1}, \hat{\Sigma}_{t-1})$, the agent treats the experience as a signal η_t^E of the true Q^* function. The posterior beliefs $(\hat{Q}_t^E, \hat{\Sigma}_t^E)$ are formed via a Bayesian update:*

$$\hat{Q}_t^E(c, s) = \hat{Q}_{t-1}(c, s) + \alpha_t^E(c, s) \underbrace{\left[\eta_t^E + \beta \hat{Q}_{t-1}(\tilde{c}_t, s_t) - \hat{Q}_{t-1}(c_{t-1}, s_{t-1})\right]}_{\textit{TD Error: } \delta_t} \tag{54}$$

where $\tilde{c}_t = \arg\max_c \hat{Q}_{t-1}(c, s_t)$ is the optimal action in the new state under current beliefs. The Kalman gain $\alpha_t^E(c, s)$ is defined by the GP covariance:

$$\alpha_t^E(c, s) = \frac{Cov(\delta_t, Q^*(c, s) \mid \mathcal{I}_{t-1})}{Var(\delta_t \mid \mathcal{I}_{t-1})} \tag{55}$$

The posterior covariance $\hat{\Sigma}_t^E$ is updated to reflect the reduction in uncertainty:

$$\hat{\Sigma}_t^E(c, s, c', s') = \hat{\Sigma}_{t-1}(c, s, c', s') - \alpha_t^E(c, s) Cov(\delta_t, Q^*(c', s') \mid \mathcal{I}_{t-1}) \tag{56}$$

Noise in Experience Signal Because the experience signal is based on a single realized transition, it is subject to noise from the stochastic transition function $\mathcal{F}$. Agents may misattribute actions to outcomes rather than to the true reason, random shocks. Examples of this are shown in the two-period and infinite-horizon examples in Appendix B.4 and Appendix B.5.

B.2.2 Learning Through Reasoning

The agent can produce mental simulations and evaluate counterfactual actions, developing a noisy vector of signals $\boldsymbol{\eta}_t^R$ that provide information about the action value function. To formalize this, fix the current state s_t and consider the vector of action values across all possible consumption levels:

$$\boldsymbol{Q}^*(s_t) = (Q^*(s_t, c_1), Q^*(s_t, c_2), \ldots, Q^*(s_t, c_n))^\top \tag{57}$$

By the GP assumption, this vector is jointly Gaussian with mean $\hat{\boldsymbol{Q}}_{t-1}(s_t)$ and covariance $\hat{\boldsymbol{\Sigma}}_{t-1}(s_t)$ derived from the kernel. We then suppose the agent can generate noisy observations of linear combinations of this vector, which represent mental simulations of different actions. The noise covariance captures the precision of the agent's mental model. The higher the precision, the more accurate the simulations, but the greater the cognitive cost. In the limit, the agent is able to perfectly evaluate all actions, equivalent to knowing Q^* exactly. This is equivalent to *model-based* learning in RL literature.

Definition 4 (Reasoning-Based Learning Update)**.** *Following the experience-based update, the agent performs internal mental simulations to refine their beliefs. At the current state s_t, the agent generates a vector of noisy reasoning signals $\boldsymbol{\eta}_t^R$, representing hypothesized action-values. These signals are modeled as:*

$$\boldsymbol{\eta}_t^R = \boldsymbol{\Omega}_t' \boldsymbol{Q}^*(s_t) + \boldsymbol{\epsilon}_{\eta,t}, \quad \boldsymbol{\epsilon}_{\eta,t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\eta,t}) \tag{58}$$

where $\boldsymbol{\Omega}_t'$ is a $k \times n$ loadings matrix chosen by the agent to determine which action comparisons are evaluated during simulation. Treating these as signals of the true Q^* function, the agent forms reasoned beliefs $(\hat{Q}_t^R, \hat{\Sigma}_t^R)$ via a Bayesian update on the intermediate beliefs $(\hat{Q}_t^E, \hat{\Sigma}_t^E)$:

$$\hat{Q}_t^R(c, s) = \hat{Q}_t^E(c, s) + \boldsymbol{\alpha}_t^R(c, s)^\top \left[\boldsymbol{\eta}_t^R - \boldsymbol{\Omega}_t' \hat{\boldsymbol{Q}}_t^E(s_t)\right] \tag{59}$$

$$\boldsymbol{\alpha}_t^R(c, s) = Cov(\boldsymbol{\eta}_t^R, Q^*(c, s) \mid \mathcal{I}_t^E) \left[Var(\boldsymbol{\eta}_t^R \mid \mathcal{I}_t^E)\right]^{-1} \tag{60}$$

The posterior covariance $\hat{\Sigma}_t^R$ is updated accordingly to reflect the reduction in uncertainty from mental simulation:

$$\hat{\Sigma}_t^R(c, s, c', s') = \hat{\Sigma}_t^E(c, s, c', s') - \boldsymbol{\alpha}_t^R(c, s)^\top Cov(\boldsymbol{\eta}_t^R, Q^*(c', s') \mid \mathcal{I}_t^E) \tag{61}$$

B.3 Action Selection

At each period t, the agent goes through the set of steps in Definition 2. They first perform a bayesian update of their beliefs based on realized utility from the previous period: *experience-based learning.* They then must choose how much reasoning to do to optimally update their beliefs: *reasoning decision.* At each step, the agent faces two tasks: one to determine optimal policy given current beliefs, and another to determine the optimal reasoning strategy to improve those beliefs.

B.3.1 The Optimal Policy Decision

The agent selects actions according to a policy $\pi_t(\cdot|s_t)$ that maximizes expected payoffs under current beliefs, subject to an exploration constraint that scales with posterior uncertainty:

$$\max_{\pi_t(\cdot|s_t)} \quad \sum_c \hat{Q}_t(c, s_t)\, \pi_t(c|s_t) \tag{62}$$

$$\text{s.t.} \quad \sum_c \pi_t(c|s_t) = 1 \tag{63}$$

$$-\sum_c \pi_t(c|s_t) \ln \pi_t(c|s_t) \geq h \cdot \mathrm{tr}\Big(\hat{\mathbf{\Sigma}}_t(s_t)\Big) \tag{64}$$

where $h > 0$ governs the strength of the exploration motive and $\mathrm{tr}(\hat{\mathbf{\Sigma}}_t(s_t)) = \sum_c \sigma_t^2(c, s_t)$ is the total posterior variance across actions. The constraint (Eq. (111)) encodes *uncertainty-driven exploration.* Standard maximum-entropy reinforcement learning imposes a fixed entropy floor. Here, the entropy bound is *endogenous*: it tightens when the agent is uncertain about action values and relaxes as learning resolves that uncertainty. As beliefs sharpen, the constraint slackens and the policy converges toward the greedy action $\arg\max_c \hat{Q}_t(c, s_t)$.

$$\max_{\pi_t(\cdot|s_t)} \quad \sum_c \hat{Q}_t(c, s_t)\, \pi_t(c|s_t) \tag{65}$$

$$\text{s.t.} \quad \sum_c \pi_t(c|s_t) = 1 \tag{66}$$

$$-\sum_c \pi_t(c|s_t) \ln \pi_t(c|s_t) \geq h \cdot \mathrm{tr}\Big(\hat{\mathbf{\Sigma}}_t(s_t)\Big) \tag{67}$$

Proposition 5 (Optimal Action Policy)**.** *The solution to the constrained maximization problem yields a state-dependent softmax distribution:*

$$\pi_t(c|s_t) = \begin{cases} \dfrac{\exp\left(\hat{Q}_t(c, s_t)/\delta_t\right)}{\sum_{c'} \exp\left(\hat{Q}_t(c', s_t)/\delta_t\right)} & \textit{if } \delta_t > 0 \\ \mathbb{I}\left(c = \arg\max_{c'} \hat{Q}_t(c', s_t)\right) & \textit{if } \delta_t = 0 \end{cases} \tag{68}$$

where $\delta_t \geq 0$ is the Lagrange multiplier associated with the entropy constraint. The multiplier δ_t acts as an endogenous temperature parameter. When uncertainty is high, δ_t is large, flattening the policy distribution.

This optimization is performed twice within the period t learning cycle. First, using the intermediate experience-based beliefs $\hat{Q}_t^E$ to form draft policy π_t^E, which is used in the reasoning process to evaluate expected payoffs from mental simulation. Second, after reasoning signals are realized, it is solved using the final reasoned beliefs $\hat{Q}_t^R$ to determine the π_t^R for action selection.

B.3.2 The Reasoning Decision

Reducing uncertainty allows the agent to tighten their priors over the action value function, and improve their policy. Because they are endowed with some reasoning capacity, they will naturally use it to improve their beliefs. However, reasoning is costly. The agents face a rational inattention problem [Sims, 2003] in which they must optimally allocate their limited cognitive resources to improve their beliefs. They choose how many simulations to run and the precision of those simulations, balancing the expected improvement in payoffs against the cognitive cost of reasoning.

The agent chooses the simulations to run ($\mathbf{\Omega}_t$) and the precision of those simulations ($\mathbf{\Sigma}_{\eta,t}$). The reasoning signal takes the form $\eta_t^R = \mathbf{\Omega}_t Q^* + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \mathbf{\Sigma}_{\eta,t})$, where rows of $\mathbf{\Omega}_t$ specify which linear combinations of $Q^*(c, s)$ to reason about, and $\mathbf{\Sigma}_{\eta,t}$ governs the noise in those comntemplations. The agent maximizes expected payoff net of reasoning costs:

$$\max_{\mathbf{\Omega}_t, \mathbf{\Sigma}_{\eta,t}} \mathbb{E}\left[\sum_c \hat{Q}_t^R(c, s_t) \cdot \pi_t(c|s_t) \mid \mathcal{I}_t^E\right] - \kappa \ln\left[\frac{|\hat{\mathbf{\Sigma}}_t^E|}{|\hat{\mathbf{\Sigma}}_t^R|}\right] \tag{69}$$

The cost term is the information gain from reasoning, measured as the entropy reduction in beliefs about Q^*. The lower the posterior covariance $\hat{\mathbf{\Sigma}}_t^R$, the more precise the beliefs, and the higher the cost; in the limit the cost of perfect information becomes infinite. The parameter κ governs the marginal cost of information.

Proposition 6 (Optimal Reasoning Allocation)**.** *Let $\hat{\mathbf{\Sigma}}_t^E = V\Lambda^E V'$ be the eigendecomposition of the prior covariance, where $\Lambda^E = \mathrm{diag}(\lambda_1^E, \ldots, \lambda_n^E)$ with $\lambda_1^E \geq \cdots \geq \lambda_n^E > 0$. The optimal posterior covariance is:*

$$\hat{\mathbf{\Sigma}}_t^R = V\Lambda^* V' \tag{70}$$

where $\Lambda^* = \mathrm{diag}(\lambda_1^*, \ldots, \lambda_n^*)$ with:

$$\lambda_i^* = \min\left\{\lambda_i^E,\ \frac{\kappa}{h\delta_t^E}\right\} \tag{71}$$

and δ_t^E is the Lagrange multiplier from the draft policy problem solved under prior beliefs $\hat{Q}_t^E$.

The optimal signal structure sets $\mathbf{\Omega}_t^* = V'$ *and* $\mathbf{\Sigma}_{\eta,t}^* = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2)$ *with:*

$$\sigma_i^2 = \begin{cases} \dfrac{\lambda_i^E \lambda_i^*}{\lambda_i^E - \lambda_i^*} & \text{if } \lambda_i^E > \dfrac{\kappa}{h\delta_t^E} \\ \infty & \text{otherwise} \end{cases} \tag{72}$$

B.4 An Example of Experience-Based Learning in a Two-Period Model

Consider a two-period model in which an agent is endowed with initial wealth s_1 and chooses consumption $c_1 \in [0, s_1]$, saving the remainder at gross interest rate R. Second-period wealth is $s_2 = R(s_1 - c_1) + \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is a mean-zero shock to the transition. The agent has quadratic utility $u(c) = c - \frac{1}{2}c^2$ and discounts at rate β. In the terminal period the agent consumes all wealth, so $c_2 = s_2$. The agent's problem is:

$$\max_{c_1 \in [0, s_1]} \left[c_1 - \frac{1}{2}c_1^2 \right] + \beta\, \mathbb{E}\left[s_2 - \frac{1}{2}s_2^2 \right] \tag{73}$$

We set $\beta R = 1$, under which certainty equivalence holds and the rational agent smooths consumption equally across periods. The optimal first-period consumption is:

$$c_1^* = \frac{R\, s_1}{1 + R} \tag{74}$$

In our decision framework this solution implies the following state-value and action-value functions:

$$\begin{aligned} Q^*(c_1, s_1) &= c_1 - \frac{1}{2}c_1^2 + \beta\left[s_2 - \frac{1}{2}s_2^2 + \sigma_\varepsilon^2 \right] \\ V^*(s_1) &= Q^*(c_1^*, s_1), \quad V^*(s_2) = s_2 - \frac{1}{2}s_2^2 \end{aligned}$$

We next consider an agent with the same utility and budget set who does not know the optimal policy or value functions. The agent holds a GP prior over Q^*:

$$Q^*(s, c) \sim GP\left(\hat{Q}_0(s, c),\ k\left((s, c), (s', c')\right) \right) \tag{75}$$

with RBF kernel:

$$k\left((s, c), (s', c')\right) = \sigma_0^2 \exp\left(-\psi_s (s - s')^2 - \psi_c (c - c')^2 \right) \tag{76}$$

We center the prior mean on the true action value function, $\hat{Q}_0 = Q^*$, so that any posterior deviation from the truth is driven entirely by noise in the experience signal rather than prior misspecification. We assume the agent chooses optimal action c_1^* in the first period, but is then subject to some transmission shock ε. Their temporal difference signal is then:

$$
\begin{aligned}
\delta &= u(c_1^*, s_1) + \beta\hat{Q}_0(s_2, s_2) - \hat{Q}_0(c_1^*, s_1) \\
&= c_1^* - \tfrac{1}{2}(c_1^*)^2 + \beta\hat{Q}_0(c_1^* + \varepsilon, c_1^* + \varepsilon) - \hat{Q}_0(c_1^*, s_1) \\
&= c_1^* - \tfrac{1}{2}(c_1^*)^2 + \beta\left[c_1^* + \varepsilon - \tfrac{1}{2}(c_1^* + \varepsilon)^2\right] - \left[c_1^* - \tfrac{1}{2}(c_1^*)^2 + \beta\mathbb{E}\left[s_2 - \tfrac{1}{2}s_2^2\right]\right] \\
&= c_1^* - \tfrac{1}{2}(c_1^*)^2 + \beta\left[c_1^* + \varepsilon - \tfrac{1}{2}(c_1^* + \varepsilon)^2\right] - c_1^* + \tfrac{1}{2}(c_1^*)^2 - \beta\left[c_1^* - \tfrac{1}{2}(c_1^*)^2 + \sigma_\varepsilon^2\right] \\
&= \beta\left[\varepsilon(1 - c_1^*) - \tfrac{1}{2}\varepsilon^2 + \tfrac{1}{2}\sigma_\varepsilon^2\right]
\end{aligned}
$$

The Kalman gain α^E is described by the covariance between the TD error δ and the action value function Q^*:

$$
\begin{aligned}
\alpha^E(s, c) &= \frac{\mathrm{Cov}(\delta, Q^*(s, c))}{\mathrm{Var}(\delta)} \\
&= \frac{\exp\left[-\psi_s(s - s_1)^2 - \psi_c(c - c_1^*)^2\right] - \beta\exp\left[-\psi_s(\varepsilon + c_1^* - s)^2 - \psi_c(\varepsilon + c_1^* - c)^2\right]}{1 + \beta^2 - 2\beta\exp\left[-\psi_c\varepsilon^2 - \psi_s\frac{(\varepsilon + \varepsilon R - s_1)^2}{(1+R)^2}\right]}
\end{aligned}
$$

Meaning for any point (s, c) in the state-action space, the agent's difference between the posterior mean $\hat{Q}^E(s, c)$ and the prior mean $\hat{Q}_0(s, c)$ is:

$$
\begin{aligned}
\hat{Q}^E(s, c) - \hat{Q}_0(s, c) &= \alpha^E(s, c) \cdot \delta \\
&= \frac{\exp\left[-\psi_s(s - s_1)^2 - \psi_c(c - c_1^*)^2\right] - \beta\exp\left[-\psi_s(\varepsilon + c_1^* - s)^2 - \psi_c(\varepsilon + c_1^* - c)^2\right]}{1 + \beta^2 - 2\beta\exp\left[-\psi_c\varepsilon^2 - \psi_s\frac{(\varepsilon + \varepsilon R - s_1)^2}{(1+R)^2}\right]} \\
&\quad \times\ \beta\left[\varepsilon(1 - c_1^*) - \tfrac{1}{2}\varepsilon^2 + \tfrac{1}{2}\sigma_\varepsilon^2\right]
\end{aligned}
$$

The difference between the posterior and prior mean at the optimal action c_1^* is contaminated by noise from the transition shock ε. Given a risk-averse agent, if ε is negative, then the agent experiences a worse outcome than expected, leading to a negative TD error and a downward update in beliefs. This is is shown in Fig. 8, which plots the posterior distortion at the period-1 state-action pair (s_1, c_1^*) against the shock ε.

The agent misattributes the shock to the action. Fig. 9 shows the Kalman gain across the state-action space for three shock realizations. Under a -2σ shock (left), the agent revises beliefs sharply downward near (s_1, c_1^*), concluding the optimal action was worse than expected. Under a $+2\sigma$ shock (right), the same action appears better than expected. Comparing the two panels, the downward revision under the negative shock is larger in magnitude than the upward revision under the equally-sized positive shock. Under no shock (center), the agent still revises slightly upward: the realized outcome at $\varepsilon = 0$ exceeds what Q^* predicted in expectation, because Q^* prices in the variance of future shocks that did not materialize,

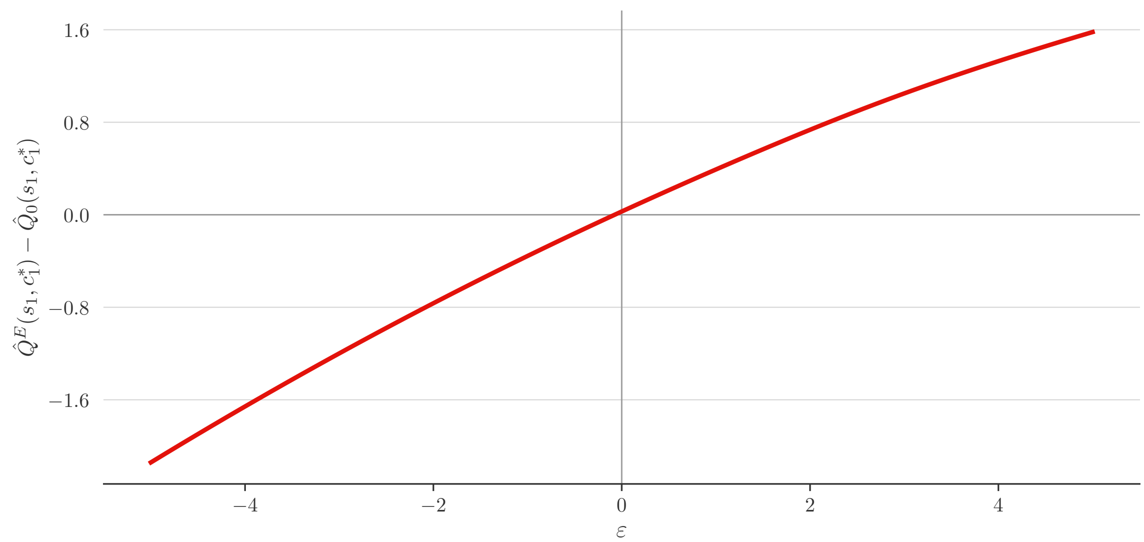

Figure 8: **Posterior Distortion at** (s_1, c_1^*) **vs. Shock.** (ε) The agent's posterior mean at starting state s_1 and optimal consumption c_1^*, $\hat{Q}^E(s_1, c_1^*)$, after the experience-based update, compared to the prior mean $\hat{Q}_0(s_1, c_1^*)$.

producing a small positive TD error of $\beta \frac{1}{2} \sigma_\varepsilon^2$.

B.5 An Example of Experience-Based Learning in Infinite-Horizon Model

We extend the two-period example to an infinite-horizon setting with adjustments for tractability. Consider an agent endowed with initial wealth s_1 who chooses a consumption path $\{c_t\}_{t=1}^{\infty}$. The agent has log utility $u(c) = \ln(c)$ and faces a multiplicative wealth transition $s_{t+1} = R(s_t - c_t) \cdot e^{\varepsilon_t}$ with $\varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ i.i.d. The agent's problem is:

$$\max_{\{c_t\}_{t=1}^{\infty}} \sum_{t=1}^{\infty} \beta^{t-1} \mathbb{E}\left[\ln(c_t)\right] \tag{77}$$

The optimal policy is $c_t^* = (1-\beta)s_t$ for all t, independent of the interest rate R. Under this policy, the true action value function and state value function are log-linear in their arguments. Defining the constant $\xi(\beta, R) \equiv \frac{\ln(1-\beta)}{1-\beta} + \frac{\beta \ln(R\beta)}{(1-\beta)^2}$:

$$Q^*(s, c) = \ln(c) + \frac{\beta}{1-\beta} \ln(R(s-c)) + \xi(\beta, R) \tag{78}$$

$$V^*(s) = \frac{1}{1-\beta} \ln s + \xi(\beta, R) \tag{79}$$

TD Error. Suppose the agent holds a GP prior centered on the truth, $\hat{Q}_0 = Q^*$, and chooses the optimal action $c_1^* = (1-\beta)s_1$ at initial wealth s_1. Upon transitioning to $s_2 = \beta R s_1 e^{\varepsilon}$, the temporal difference reduces to a simple function of the shock ε.

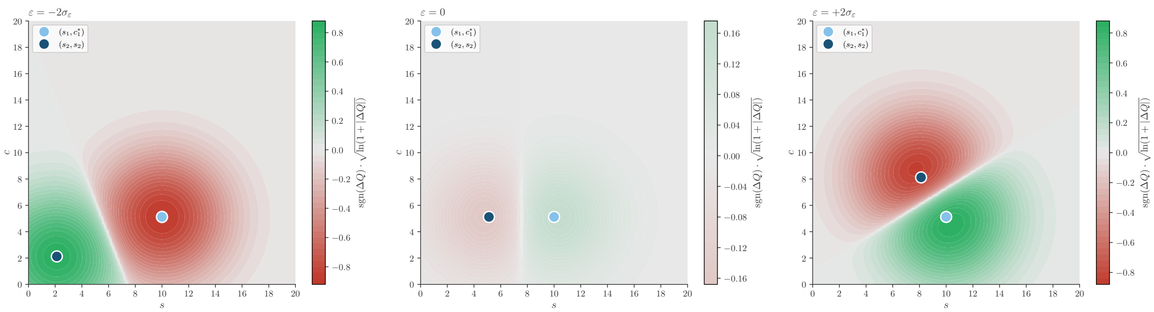

Figure 9: **Posterior Distortion Surface.** Kalman gain $\alpha^E(s, c)$ across the state-action space for three shock realizations. Green indicates upward revision, red indicates downward. The light blue marker denotes the period-1 state-action pair (s_1, c_1^*); the dark blue marker denotes the period-2 point (s_2, s_2). Colors are displayed on a $\text{sgn}(\cdot)\sqrt{\ln(1 + |\cdot|)}$ scale, which linearizes the RBF kernel's exponential-quadratic decay in distance, so that the color gradient is approximately proportional to distance from the observation points. Panel titles report the raw error δ; the full posterior update at any point is $\alpha^E(s, c) \cdot \delta$.

$$\delta = \frac{\beta}{1-\beta}\left[\ln(\beta R s_1) - \ln(\beta R s_1 e^{\varepsilon})\right] = -\frac{\beta}{1-\beta}\,\varepsilon \tag{80}$$

The temporal difference is linear in the shock with no higher-order terms. Its variance is $\text{Var}(\delta) = \left(\frac{\beta}{1-\beta}\right)^2 \sigma_\varepsilon^2$, scaling with the squared infinite-horizon discount multiplier. Because the optimal policy and value functions are stationary, this result holds at any period: an agent at state s_{t-1} who takes optimal action $c_{t-1}^* = (1-\beta)s_{t-1}$ and transitions to $s_t = \beta R s_{t-1} e^{\varepsilon_t}$ experiences the same temporal difference $\delta_t = -\frac{\beta}{1-\beta}\varepsilon_t$, independent of wealth.

Posterior Update. To respect the multiplicative nature of the wealth transition, the agent's similarity measure must be scale-invariant. The natural choice is the *Log-Laplace kernel.*

$$k\big((s, c), (s', c')\big) = \sigma_0^2 \exp\left(-\psi_s \left|\ln \frac{s}{s'}\right| - \psi_c \left|\ln \frac{c}{c'}\right|\right) \tag{81}$$

The form captures the correct geometry, but the absolute values require piecewise integration. However, we can derive a global closed-form solution by exploiting the identity $\exp(-|z|) = \min(e^z, e^{-z})$. Applied to log-ratios, this transforms the kernel into the product of minimum ratios.

$$\exp\left(-\psi \left|\ln \frac{x}{y}\right|\right) = \left[\min\left(\frac{x}{y}, \frac{y}{x}\right)\right]^{\psi} \tag{82}$$

This allows us to evaluate the kernel covariance analytically across the entire state space. Let $\rho(\varepsilon)$ denote the realized growth factor of wealth from $t-1$ to period t. Substituting the optimal policy into the transition equation $s_t = R(s_{t-1} - c_{t-1}^*)e^{\varepsilon}$ yields:

$$\rho(\varepsilon) \equiv \frac{s_t}{s_{t-1}} = \frac{c_t^*}{c_{t-1}^*} = \beta R e^{\varepsilon} \tag{83}$$

Reiterating that the agent's prior is centered on the truth, the decision point is (s_{t-1}, c^*_{t-1}) and the outcome point is $(s_t, c^*_t) = (s_{t-1}\rho(\varepsilon), c^*_{t-1}\rho(\varepsilon))$. Using the simplified min-ratio form, the update is:

$$\hat{Q}^E(s,c) - \hat{Q}_0(s,c) = \left(-\frac{\beta\varepsilon}{1-\beta}\right) \cdot \frac{\mathcal{K}_{t-1}(s,c) - \beta\mathcal{K}_t(s,c,\varepsilon)}{1+\beta^2 - 2\beta\left[\min\left(\rho(\varepsilon), \rho(\varepsilon)^{-1}\right)\right]^{\psi_s+\psi_c}} \tag{84}$$

where the spatial decay terms $\mathcal{K}_{t-1}$ (anchored at the decision) and $\mathcal{K}_t$ (anchored at the outcome) are given by:

$$\mathcal{K}_{t-1}(s,c) = \left[\min\left(\frac{s}{s_{t-1}}, \frac{s_{t-1}}{s}\right)\right]^{\psi_s} \left[\min\left(\frac{c}{c^*_{t-1}}, \frac{c^*_{t-1}}{c}\right)\right]^{\psi_c} \tag{85}$$

$$\mathcal{K}_t(s,c,\varepsilon) = \left[\min\left(\frac{s}{s_t}, \frac{s_t}{s}\right)\right]^{\psi_s} \left[\min\left(\frac{c}{c^*_t}, \frac{c^*_t}{c}\right)\right]^{\psi_c} \tag{86}$$

Kernel Evaluations. Let $\psi \equiv \psi_s + \psi_c$ and define the *min-ratio kernel overlap*:

$$m(\varepsilon) \equiv \left[\min\big(\rho(\varepsilon),\ \rho(\varepsilon)^{-1}\big)\right]^{\psi} \tag{87}$$

where $\rho(\varepsilon) = \beta R e^{\varepsilon}$. Both savings and consumption scale by the common factor $\rho(\varepsilon)$. Each kernel evaluation reduces to a power of ρ alone.

$$\mathcal{K}_{t-1}(s_{t-1}, c^*_{t-1}) = 1, \qquad \mathcal{K}_t(s_{t-1}, c^*_{t-1}, \varepsilon) = m(\varepsilon),$$
$$\mathcal{K}_{t-1}(s_t, c^*_t) = m(\varepsilon), \qquad \mathcal{K}_t(s_t, c^*_t, \varepsilon) = 1,$$

$$\hat{Q}^E(s_{t-1}, c^*_{t-1}) - \hat{Q}_0(s_{t-1}, c^*_{t-1}) = \frac{-\beta\varepsilon}{(1-\beta)} \cdot \frac{1 - \beta\, m(\varepsilon)}{1+\beta^2 - 2\beta\, m(\varepsilon)}, \tag{88}$$

$$\hat{Q}^E(s_t, c^*_t) - \hat{Q}_0(s_t, c^*_t) = \frac{-\beta\varepsilon}{(1-\beta)} \cdot \frac{m(\varepsilon) - \beta}{1+\beta^2 - 2\beta\, m(\varepsilon)}. \tag{89}$$

The two updates share a common denominator and prefactor.

$$m(\varepsilon) = \begin{cases} (\beta R e^{\varepsilon})^{\psi} & \text{if } \varepsilon < \varepsilon^* \quad (\rho < 1), \\ (\beta R e^{\varepsilon})^{-\psi} & \text{if } \varepsilon \geq \varepsilon^* \quad (\rho \geq 1). \end{cases} \tag{90}$$

B.6 An Exstension to Allow Caution in Experience-Based Learning

In the experience-based update, Definition 6, the decision-maker treats her realized utility η^E_t as if it perfectly identifies the Bellman residual $Q^*(c_{t-1}, s_{t-1}) - \beta Q^*(\tilde{c}_t, s_t)$. The interpretation is that the agent is naive to the underlying stochastic transitions and has no knowledge of the transition distribution $\mathcal{F}$. Here we propose a small generalization of the experience-based update in which the agent adds noise to her experience signal. This added noise admits two interpretations. First, the agent may be partially aware that she resides within a stochastic environment, and cautiously discounts her experience signal to reflect the fact that any single realized transition may be unrepresentative of the expectation it proxies for. Second, the agent may not perfectly observe the realized utility η_t^E itself, instead receiving only a noisy signal of it.

$$\eta_t^E = Q^*(c_{t-1}, s_{t-1}) - \beta Q^*(\tilde{c}_t, s_t) + \varepsilon_t^C, \qquad \varepsilon_t^C \sim \mathcal{N}(0, \sigma_C^2) \tag{91}$$

where $\sigma_C^2 \geq 0$ parameterizes the agent's confidence in any single experience and ε_t^C is independent across periods and independent of Q^*.

Definition 5 (Cautious Empirical Learning Update)**.** *Upon arriving at period t, the agent observes realized utility $\eta_t^E = u(c_{t-1}, s_{t-1})$ and the transition to state s_t. Given prior beliefs $(\hat{Q}_{t-1}, \hat{\Sigma}_{t-1})$ and signal noise σ_C^2, the posterior beliefs $(\hat{Q}_t^E, \hat{\Sigma}_t^E)$ are:*

$$\hat{Q}_t^E(c, s) = \hat{Q}_{t-1}(c, s) + \alpha_t^E(c, s) \cdot \delta_t \tag{92}$$

where $\delta_t = \eta_t^E + \beta \hat{Q}_{t-1}(\tilde{c}_t, s_t) - \hat{Q}_{t-1}(c_{t-1}, s_{t-1})$ and $\tilde{c}_t = \arg\max_c \hat{Q}_{t-1}(c, s_t)$. The Kalman gain is:

$$\alpha_t^E(c, s) = \frac{\mathrm{Cov}(\delta_t, Q^*(c, s) \mid \mathcal{I}_{t-1})}{\mathrm{Var}(\delta_t \mid \mathcal{I}_{t-1}) + \sigma_C^2} \tag{93}$$

The posterior covariance is:

$$\hat{\Sigma}_t^E(c, s, c', s') = \hat{\Sigma}_{t-1}(c, s, c', s') - \alpha_t^E(c, s) \cdot \mathrm{Cov}(\delta_t, Q^*(c', s') \mid \mathcal{I}_{t-1}) \tag{94}$$

The only modification relative to Definition 6 is the additive σ_C^2 in the denominator of the Kalman gain.

$$\sigma_C^2 \begin{cases} \in \big[0,\ \beta^2 \mathrm{Var}(V^*(s') \mid s, c)\big) & \text{credulous} \\ = \beta^2 \mathrm{Var}(V^*(s') \mid s, c) & \text{rational Bayesian} \\ > \beta^2 \mathrm{Var}(V^*(s') \mid s, c) & \text{cautious} \end{cases} \tag{95}$$

When $\sigma_C^2 = 0$, the agent trusts each experience fully and Definition 5 reduces exactly to Definition 6. When $\sigma_C^2 = \beta^2 \mathrm{Var}(V^*(s') \mid s, c)$, the agent correctly accounts for the sampling variance inherent in replacing the Bellman expectation with a single realized transition. When $\sigma_C^2 > \beta^2 \mathrm{Var}(V^*(s') \mid s, c)$, the agent is overly cautious, discounting her experience signal more than is warranted by the true sampling variance.

B.7 An Extension to Allow for Information-Directed Policies

The optimal policy in Appendix B.3 imposes an entropy constraint that scales with posterior uncertainty. This is a reduced-form device for ensuring that the agent values information acquisition, not just the discounted stream of payoffs implied by her current beliefs. Naturally, a consumer or firm benefits from resolving uncertainty about Q^*, because doing so allows them to refine their policy in all future periods. The entropy constraint encodes this motive partially by ensuring the agent explores more when uncertainty is high, but it does not capture the full value of information. How might we formalize in a decision-theoretic way? In fact, the true value function the decision-maker faces is not simply a function of the current state, but also of her beliefs $(\hat{Q}_t, \hat{\Sigma}_t)$ about the action-value function.

$$V^*(s_t, \hat{Q}_t, \hat{\Sigma}_t) = \max_{\pi_t} \left\{ \int \hat{Q}_t(c, s_t)\, \pi_t(c)\, dc + \beta\, \mathbb{E}\Big[V^*(s_{t+1}, \hat{Q}_{t+1}, \hat{\Sigma}_{t+1}) \;\Big|\; \pi_t \Big] \right\} \tag{96}$$

This is a Bayes-adaptive MDP (BAMPD) in which the agent's beliefs are part of the state space Duff [2001]. Rather than solving the BAMDP directly, we might formalize the decision-maker's problem as in Russo and Roy [2014, 2017], using Information-Directed Sampling (IDS) to minimize the information-ratio, the squared expected regret per unit of information gained about the optimal action.

$$\pi_t^{\text{IDS}} = \arg \min_{\pi_t \in \mathcal{D}(B(s_t))} \frac{\Delta_t(\pi_t)^2}{g_t(\pi_t)} \tag{97}$$

where $\Delta_t(\pi_t)$ is the expected instantaneous regret and $g_t(\pi_t)$ is the mutual information between the optimal action and the next TD observation. This objective is chosen given Russo and Roy [2017]: the expected cumulative regret of any policy satisfies $\mathbb{E}[\text{Regret}(T)] \leq \sqrt{\bar{\Psi} \cdot H(\alpha_1) \cdot T}$, where $H(\alpha_1)$ is the prior entropy over the optimal action and $\bar{\Psi}$ bounds the per-period information ratio, so that minimizing Ψ_t each period minimizes the tightest available bound on lifetime regret. However, there are two challenges to using the IDS approach in our setting.

Continuous action spaces. The IDS framework is developed for finite action sets, where the entropy $H(\alpha_1)$ of the prior over the optimal action is Shannon entropy. With continuous consumption $c \in B(s_t)$, the optimal action $c^* = \arg\max_c Q^*(c, s_t)$ is a continuous random variable. For continuous random variables, the analog of Shannon entropy is differential entropy, which unlike its discrete counterpart can be negative and is not invariant to reparameterization of the underlying variable. In addition, the regret bound does not extend directly to our setting. Subsequently, there is then no clear motivation for minimizing the information ratio. These issues are partially mitigated by substituting differential information with mutual information $I_t(c^*; \delta_{t+1})$ in the denominator, which is invariant to reparameterization, but the regret bound still does not hold.

Transition dynamics. In our current model, the agent is completely oblivious to the transition dynamics. This makes the agent unable to calculate expected temporal difference (δ_{t+1}) from any action c_t, which is necessary to evaluate mutual information $I_t(c^*; \delta_{t+1})$. To allow for IDS policies, the agent needs to be endowed with some knowledge of the transition dynamics, so that they can calculate information-gain and payoff tradeoffs from different actions.

B.7.1 Extending by Allowing a GP Prior over Transition Dynamics

We can extend the framework to allow the decision maker to have *model uncertainty*: she is uncertain not only about the action value function Q^*, but also about the dynamics of the environment in which she operates. Motivated by Deisenroth and Rasmussen [2011], we endow the agent with a second GP prior, this time over the transition function $\mathcal{F}$, so that the agent maintains probabilistic beliefs about how her actions map to future states alongside her beliefs about action values. She updates her beliefs about $\mathcal{F}$ using a standard Bayesian update on the observed transition. Upon arriving at state s_{t+1} after taking action c_t in state s_t, the agent perfectly observes the realized state change $\Delta_{t+1} = s_{t+1} - s_t$. She treats this as an exact observation of $\mathcal{F}$ at the input (s_t, c_t):

$$\hat{f}_{t+1}(s,c) = \hat{f}_t(s,c) + \alpha_t^f(s,c) \cdot \left[\Delta_{t+1} - \hat{f}_t(s_t, c_t)\right] \tag{98}$$

$$\alpha_t^f(s,c) = \frac{\hat{\Sigma}_t^f((s_t,c_t),(s,c))}{\hat{\Sigma}_t^f((s_t,c_t),(s_t,c_t))} \tag{99}$$

As with the experience-based update of Q^*, the agent attributes the entire prediction error to signal rather than noise: she does not account for the stochastic shock ϵ_{t+1} embedded in the realized transition.

IDS. The agent can forecast the distribution of her TD observation δ_{t+1} before acting, which makes the IDS objective $\Delta_t(\pi)^2/g_t(\pi)$ computable. She can now evaluate how informative each action is about which consumption level is truly optimal, and direct her exploration according to the information ratio. However, mutual information remains an unnatural preference object, and the regret bound that motivates minimizing the information ratio in the finite-action setting does not extend to our continuous consumption problem.

Information Shadow Price. Another possible formulation gives information gain a price. The agent maximizes expected payoff plus a premium for decision-relevant information:

$$c_t^* = \arg\max_{c \in B(\mathbf{s}_t)} \left[\hat{Q}_t(c, \mathbf{s}_t) + \iota_t \cdot g_t(c)\right] \tag{100}$$

where $\iota_t \geq 0$ is the shadow cost of ignorance, governing the agent's willingness to sacrifice immediate payoff for information that improves future decisions. In the fully optimal BAMDP, ι_t would emerge endogenously from the continuation value; here it is treated as a parameter of the agent's preference.

Definition 6 (Experience-Based Learning Update)**.** *Upon arriving at period t, the firm observes the realized dividend $d_{t-1} = z_{t-1}k_{t-1}^{\alpha} - i_{t-1} - \frac{\phi}{2}\frac{i_{t-1}^2}{k_{t-1}} + b_t - (1+R)b_{t-1}$ and the transition to state (z_t, k_t, b_t). Given the prior beliefs $(\hat{Q}_{t-1}, \hat{\Sigma}_{t-1})$, the firm defines the experience-based signal*

$$\eta_t^E \equiv d_{t-1} \approx Q^*(z_{t-1}, k_{t-1}, i_{t-1}) - \beta\, Q^*\big(\tilde{i}_t, z_t, k_t\big) \tag{101}$$

where $\tilde{i}_t = \arg\max_i \hat{Q}_{t-1}(z_t, k_t, i)$ is the greedy action under current beliefs. The posterior mean is updated via

$$\hat{Q}_t^E(z,k,i) = \hat{Q}_{t-1}(z,k,i) + \alpha_t^E(z,k,i) \underbrace{\Big[\eta_t^E - \hat{Q}_{t-1}(z_{t-1}, k_{t-1}, i_{t-1}) + \beta\, \hat{Q}_{t-1}(z_t, k_t, \tilde{i}_t)\Big]}_{\text{TD surprise: } \delta_t} \tag{102}$$

where the Kalman gain α_t^E is determined by the GP covariance structure:

$$\alpha_t^E(z,k,i) = \frac{\operatorname{Cov}\big(\eta_t^E,\, Q^*(z,k,i) \mid \mathcal{I}_{t-1}\big)}{\operatorname{Var}\big(\eta_t^E \mid \mathcal{I}_{t-1}\big)} \tag{103}$$

The posterior covariance is reduced accordingly:

$$\hat{\Sigma}_t^E\big((z,k,i),\, (z',k',i')\big) = \hat{\Sigma}_{t-1}\big((z,k,i),\, (z',k',i')\big) - \alpha_t^E(z,k,i)\operatorname{Cov}\big(\eta_t^E,\, Q^*(z',k',i') \mid \mathcal{I}_{t-1}\big) \tag{104}$$

Definition 7 (Reasoning-Based Learning Update)**.** *Following the experience-based update, the firm engages in costly mental simulation to refine its beliefs about Q^* at the current state (z_t, k_t). The firm generates a vector of reasoning signals $\boldsymbol{\eta}_t^R$ over the feasible set of investment actions:*

$$\boldsymbol{\eta}_t^R = \boldsymbol{\Omega}_t' \boldsymbol{Q}^*(z_t, k_t, \cdot) + \boldsymbol{\varepsilon}_{\eta,t}, \quad \boldsymbol{\varepsilon}_{\eta,t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\eta,t}) \tag{105}$$

where $\boldsymbol{\Omega}_t'$ is a loadings matrix chosen by the firm to determine which investment comparisons are evaluated during deliberation. The posterior beliefs are updated via

$$\hat{Q}_t^R(z,k,i) = \hat{Q}_t^E(z,k,i) + \boldsymbol{\alpha}_t^R(z,k,i)^\top \Big[\boldsymbol{\eta}_t^R - \boldsymbol{\Omega}_t' \hat{\boldsymbol{Q}}_t^E(z_t, k_t, \cdot)\Big] \tag{106}$$

$$\boldsymbol{\alpha}_t^R(z,k,i) = \operatorname{Cov}\big(\boldsymbol{\eta}_t^R,\, Q^*(z,k,i) \mid \mathcal{I}_t^E\big)\left[\operatorname{Var}\big(\boldsymbol{\eta}_t^R \mid \mathcal{I}_t^E\big)\right]^{-1} \tag{107}$$

The posterior covariance is reduced by the information content of the reasoning signals:

$$\hat{\Sigma}_t^R\big((z,k,i),\, (z',k',i')\big) = \hat{\Sigma}_t^E\big((z,k,i),\, (z',k',i')\big) - \boldsymbol{\alpha}_t^R(z,k,i)^\top \operatorname{Cov}\big(\boldsymbol{\eta}_t^R,\, Q^*(z',k',i') \mid \mathcal{I}_t^E\big) \tag{108}$$

Proposition 7 (Optimal Investment Policy)**.** *Given end-of-period beliefs* $(\hat{Q}_t, \hat{\Sigma}_t)$*, the firm chooses an investment policy* $\pi_t(i \mid z_t, k_t)$ *to maximize expected value subject to an entropy constraint that ensures exploration proportional to remaining uncertainty:*

$$\max_{\pi_t(\cdot|z_t,k_t)} \quad \sum_{i \in \mathcal{I}(z_t,k_t)} \hat{Q}_t(z_t, k_t, i)\, \pi_t(i \mid z_t, k_t) \tag{109}$$

$$s.t. \quad \sum_i \pi_t(i \mid z_t, k_t) = 1 \tag{110}$$

$$-\sum_i \pi_t(i \mid z_t, k_t) \ln \pi_t(i \mid z_t, k_t) \geq h \cdot \mathrm{tr}\Big(\hat{\mathbf{\Sigma}}_t(z_t, k_t)\Big) \tag{111}$$

where $\mathcal{I}(z_t, k_t)$ *is the set of feasible investment levels satisfying* $d_t \geq 0$*. The solution is a state-dependent softmax distribution:*

$$\pi_t(i \mid z_t, k_t) = \begin{cases} \dfrac{\exp\Big(\hat{Q}_t(z_t, k_t, i) \,/\, \delta_t\Big)}{\sum_{i'} \exp\Big(\hat{Q}_t(z_t, k_t, i') \,/\, \delta_t\Big)} & \text{if } \delta_t > 0 \\ \mathbb{I}\Big(i = \arg\max_{i'} \hat{Q}_t(z_t, k_t, i')\Big) & \text{if } \delta_t = 0 \end{cases} \tag{112}$$

where $\delta_t \geq 0$ *is the Lagrange multiplier on the entropy constraint. The multiplier acts as an endogenous temperature: when posterior uncertainty over* Q^* *is high,* δ_t *is large and the firm spreads probability across many investment levels; as uncertainty resolves,* $\delta_t \to 0$ *and the policy concentrates on the greedy action.*

C Proofs

The following section contains the proofs of all propositions stated in the main text.

C.1 Proof of Proposition 2

Proof. Starting from the firm's original bellman equation $\mathbf{s_t} = (z, k, b)$, including the presence of debt, we have: $V(z, k, b) = \max_{k',b'}\{d(z, k, k', b, b') + \beta\, \mathbb{E}_{z'}[V(z', k', b')]\}$, subject to $b' \leq \theta k'$. We then conjecture that the value function is additively separable: $V(z, k, b) = V^K(z, k) + V^B(b)$, where $V^B(b) = -(1 + R)b$. Substituting into the original Bellman equation yields:

$$V^K(z,k) - (1+R)b = \max_{k',b'} \left\{ zk^\alpha - k' + b' - (1+R)b + \beta \mathbb{E}_{z'} \left[V^K(z', k') - (1+R)b' \right] \right\} \tag{113}$$

$$V^K(z,k) = \max_{k',b'} \left\{ zk^\alpha - k' + b' + \beta \mathbb{E}_{z'} \left[V^K(z', k') \right] - \beta(1+R)b' \right\} \tag{114}$$

$$V^K(z,k) = \max_{k',b'} \left\{ zk^\alpha - k' + b'[1 - \beta(1+R)] + \beta \mathbb{E}_{z'} \left[V^K(z', k') \right] \right\} \tag{115}$$

Since b' enters linearly, the solution is $b' = \theta k'$ when $\beta(1+R) \leq 1$ and $b' = 0$ otherwise, verifying the conjecture. ■

C.2 Proof of Proposition 4

Proof. Let Σ_E and Σ_R denote the covariance kernel functions of the experience update and the reasoning mechanism, respectively. By Mercer's theorem, the experience kernel can be expressed as an infinite sum of eigenfunctions and eigenvalues:

$$\Sigma_E(x, x') = \sum_{i=1}^{\infty} \lambda_i^E \phi_i^E(x) \phi_i^E(x') \tag{116}$$

Where $\{\lambda_i^E\}_{i=1}^{\infty}$ are the eigenvalues of the experience kernel, and $\{\phi_i^E\}_{i=1}^{\infty}$ are the corresponding orthonormal eigenfunctions. Since Σ_E is a trace-class operator, we have $\mathrm{Tr}(\Sigma_E) = \sum_{i=1}^{\infty} \lambda_i^E < \infty$. By the reasoning-based update definition (Definition 7), there exists a direct relationship between the eigenvalues of the experience update and the reasoning mechanism:

$$\lambda_i^R = \min\left\{\lambda_i^E, \frac{\kappa}{H \cdot \delta_E}\right\}, \quad \forall i. \tag{117}$$

This implies that the reasoning mechanism's covariance kernel can be expressed as:

$$\Sigma_R(x, x') = \sum_{i=1}^{\infty} \min\left\{\lambda_i^E, \frac{\kappa}{H \cdot \delta_E}\right\} \phi_i^E(x) \phi_i^E(x'). \tag{118}$$

Which, in turn, also implies $\mathrm{Tr}(\Sigma_R) \leq \mathrm{Tr}(\Sigma_E) < \infty$ for any $\kappa > 0$. Consequently, the total posterior uncertainty is given by the trace of the operator:

$$\mathrm{Tr}(\Sigma_R) = \sum_{i=1}^{\infty} \min\left\{\lambda_i^E, \frac{\kappa}{H \cdot \delta_E}\right\} < \infty. \tag{119}$$

Taking the derivative of the total posterior uncertainty with respect to the reasoning cost κ yields:

$$\frac{\partial \mathrm{Tr}(\Sigma_R)}{\partial \kappa} = \frac{1}{H \cdot \delta_E} \sum_{i=1}^{\infty} \mathbb{1}_{\left\{\lambda_i^E > \frac{\kappa}{H \cdot \delta_E}\right\}} \geq 0. \tag{120}$$

This implies that for any $\kappa_2 > \kappa_1 > 0$, we have $\mathrm{Tr}(\Sigma_R(\kappa_2)) \geq \mathrm{Tr}(\Sigma_R(\kappa_1))$. Since the expected payoff $\mathbb{E}_\pi[\hat{Q}(c, s_t)]$ is strictly increasing in policy precision, the entropy constraint $\mathcal{H}(\pi_t) \geq H \cdot \mathrm{Tr}(\Sigma_R)$ is locally non-satiated. By complementary slackness, the constraint binds at the optimum, and the exploration floor is attained:

$$\underline{\mathcal{H}}(\kappa) \equiv \inf_{\pi_t} \mathcal{H}(\pi_t) = H \cdot \mathrm{Tr}(\Sigma_R) = H \sum_{i=1}^{\infty} \min\left\{\lambda_i^E, \frac{\kappa}{H \cdot \delta_E}\right\} > 0. \tag{121}$$

Taking the partial derivative with respect to the cognitive cost κ yields:

$$\frac{\partial \underline{\mathcal{H}}}{\partial \kappa} = \frac{1}{\delta_E} \sum_{i=1}^{\infty} \nVdash_{\left\{\lambda_i^E > \frac{\kappa}{H \cdot \delta_E}\right\}} > 0. \tag{122}$$

The strict positivity holds for any $\kappa < \bar{\kappa} = H \cdot \delta_E \cdot \sup_i \lambda_i^E$, since at least one eigenvalue exceeds the water level $\kappa/(H \cdot \delta_E)$. It follows that for any $\bar{\kappa} > \kappa_2 > \kappa_1 > 0$:

$$\underline{\mathcal{H}}(\kappa_2) > \underline{\mathcal{H}}(\kappa_1), \tag{123}$$

establishing that the minimum achievable policy entropy is strictly increasing in reasoning costs. ■

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